

THEORIES OF CHEMICAL BONDING

3.1 Concept of Covalent Bond Formation

The formation of a covalent bond is due to sharing of electrons, between atoms of two elements. In order to understand it, let us look at the H – H bond in the H_2 molecule. As two hydrogen atoms come closer together, **electrostatic interactions** begin to develop between them. The two positively charged nuclei repel each other and the two negatively charged electrons also repel each other. Each nucleus attracts both electrons Figure (3.1). To understand the formation of H_2 molecule, the Fig. (3.1) has certain parameters. $A^{\oplus}$ and $B^{\oplus}$ are two nuclei of two H-atoms. Two electrons are as 1 and 2. The internuclear distance is r_{AB} and between two electrons is r_{12} . Electron 1, is at distances r_{A1} and r_{B1} from nuclei A and B respectively. The electron 2 is at r_{A2} and r_{B2} from them. There are four attractive forces. If the attractive forces are stronger than the repulsive forces, a covalent bond is formed. In this way the two atoms join together and the two shared electrons occupy the region between the nuclei.

The shared electrons act as a kind of “glue” to bind the two nuclei together into an H_2 molecule.

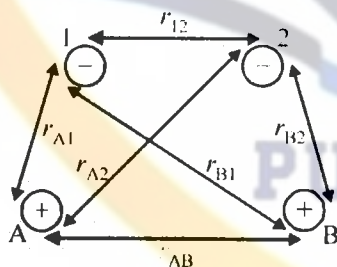


Fig. (3.1): The hydrogen molecule H_2 , consisting of two protons and two electrons.

Both nuclei are **simultaneously attracted** to the same electrons and are therefore held together.

The values of the various attractive and repulsive forces between nuclei and electrons in a covalent bond depend on how close the atoms are. If the hydrogen atoms are too far apart, the attractive forces are small and no bond exists. If the hydrogen atoms are too close together, the repulsive interaction between the nuclei becomes so strong that it pushes the atoms apart.

There is an **optimum distance** between nuclei called the **bond length** where net attractive forces are **maximized**. Here the H – H molecule is most stable Fig. (3.2). In the H_2 molecule, the bond length is 74 pm. On a graph of energy versus inter-nuclear distance, the **bond length corresponds to the minimum-energy**. This is the situation of most stable arrangement.

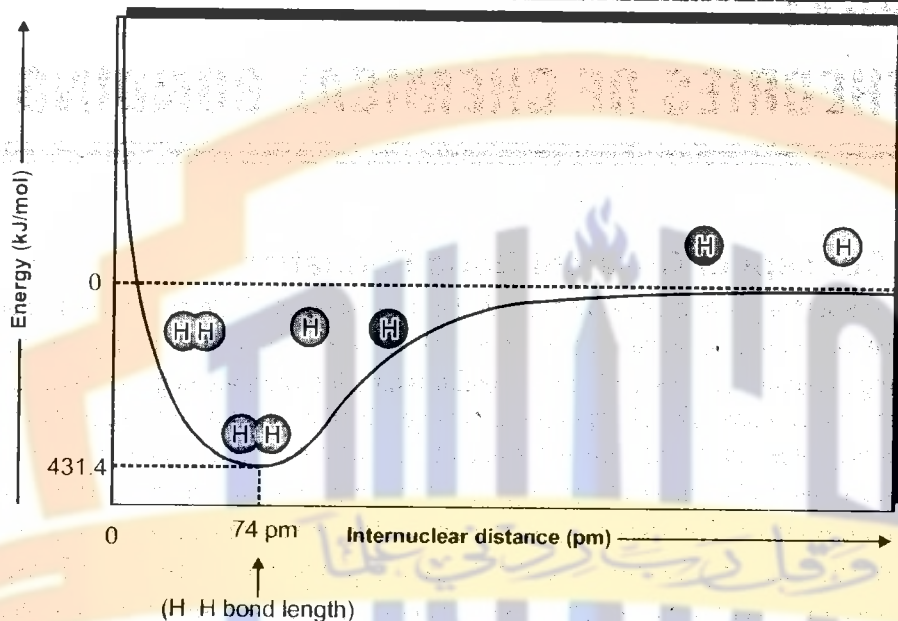


Fig. (3.2): Potential energy profile as a function of the distance between the nuclei of the hydrogen atoms. As the atoms approach each other (right side of graph), the energy decreases until the distance reaches to bond distance and then begins to increase again due to repulsions.

THEORIES OF CHEMICAL BONDING

The concept of covalent bond tells us that every covalent bond consists of a pair of electrons which hold two nuclei together. The hydrogen molecule H_2 is the simplest molecule in which there is an electron-pair bond. There are two basic quantum-mechanical treatments of the hydrogen molecule.

- (i) One of them, the valence-bond method. It considers the electron-pair bond and constructs a wave function for it by first taking the products of two atomic wave functions, one for each electron.
- (ii) The other treatment, is the molecular-orbital (MO) method. It is primarily concerned with the orbitals for single electrons. Molecular orbitals are often constructed as linear combinations of atomic orbitals (LCAO). The electrons are fed into the corresponding orbitals with due regard to the Pauli principle.

3.2 Valence Bond Theory (VBT); Quantum Mechanical Treatment

This theory was given by L. Pauling. The main points of valence bond theory are as follows.

1. Atoms retain their individuality in the molecule.
2. The bond is formed due to the interaction of valence electrons as the atoms approach each other. The inner atomic orbits from each atom not forming the bond remain undisturbed.
3. When atoms form a bond, only the orbital having valence electrons from each bonded atom lose their identity.

4. The stability of the bond is due to the fact that electrons forming the bond between two atoms undergo **exchange** between them.

Before going into the details of this theory, we should understand the concept of wave function of electron in an orbital of an atom. The wave functions of various orbitals, which have certain electrons in an atom have their mathematical expressions. These expressions are obtained from solution of Schrödinger wave equation for H-atom. The expression of ψ_{1s} is $\frac{1}{\sqrt{\pi}}$

$\left(\frac{Z}{a_0}\right)^{3/2} e^{-\frac{Zr}{a_0}}$ and ψ_{2s} is $\frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0}\right)^{3/2} \left[2 - \frac{Zr}{a_0}\right] e^{-\frac{Zr}{2a_0}}$. The wave functions of 1s and 2s are independent of θ and ϕ . Angle θ is called azimuthal angle and ϕ is called zenith angle.

Theorems of Valence Bond Theory:

There are two **theorems** on which VBT is based:

1. If Ψ_a and Ψ_b are the **wave functions** for two isolated independent atoms, with energies E_a and E_b , respectively, then the total wave function ψ of the system can be written as a product of the wave functions of the two atoms.

$$\Psi = \Psi_a \Psi_b \quad (i)$$

and energy of the system by

$$E = E_a \pm E_b \quad (ii)$$

2. If a system can be represented by a number of wave functions such as $\Psi_1, \Psi_2, \Psi_3, \dots$, then the true wave function or actual wave function, Ψ can be obtained by taking a linear combination of all these wave functions.

$$\Psi = N (C_1 \Psi_1 + C_2 \Psi_2 + C_3 \Psi_3 + \dots) \quad (iii)$$

where N is the **normalization constant** and $C_1, C_2, C_3, \dots$, are the **coefficients** indicating the weight of each of the Ψ_n 's. They are so adjusted as to give a state of lowest energy. The squares of the coefficients $C_1^2, C_2^2, \dots$ may be taken as a measure of the weights of each wave function to the total wave function.

Keeping these theorems in view, we are going to discuss the formation of H_2 molecule from two H-atoms.

Formation of the Hydrogen Molecule:

Now let us apply the concept of valence bond theory (**VBT**) of the formation of hydrogen molecule from hydrogen atoms. For this we have to consider the various forces of attraction and repulsion which may come into play. When two hydrogen atoms combine to make a covalent bond then amount of energy evolved is $431.4 \text{ kJ mol}^{-1}$. This is bond formation energy. This is the amount of **energy lowered** and is experimental value.

Now, we try to construct the wave functions of H_2 from two wave function of H-atoms. The new wave functions are also called **trial wave functions**.

When two hydrogen atoms are infinitely apart then there is no interaction between them.

$$\Psi = \Psi_a \Psi_b \quad (iv)$$

Let Ψ_a and Ψ_b represent the wave functions for two hydrogen atoms of H_a and H_b respectively.

Now, if we bring two H-atoms close together, at a distance close enough for interaction, the new wave function with wave function Ψ_1 .

$$\Psi_1 = \Psi_a(1) \cdot \Psi_b(2) \quad (v)$$

The newly constructed wave function Ψ_1 of H_2 molecule is represented in Fig (3.3). H_a and H_b are two hydrogen atoms and the electrons as 1 and 2. $\Psi_a(1)$ means that electron '1' of atom H_a revolves around its nucleus. $\Psi_b(2)$ means that electron '2' of H_b revolves around its own nucleus. The mathematical expressions of Ψ_a and Ψ_b can be obtained from solution of Schrödinger wave equation.

The wave function Ψ_1 , the **first trial wave function**, is the combined wave function of two atoms as given by equation (v). The energy of the system consisting of two hydrogen atoms may be calculated. For this purpose we need internuclear distance r_{ab} by applying **Schrödinger wave equation**. The form of Schrödinger wave equation used at Fig. (3.3) at this stage is

$$E = \frac{\int \Psi H \Psi d\tau}{\int \Psi^2 d\tau} \quad (vi)$$

H = **Hamiltonian operator** (It involves **Laplacian operator** ∇^2 . Please consult Schrödinger wave equation for this purpose)

$d\tau$ = Volume element of electronic cloud

Ψ_1 = Resulting wave function which we have constructed in equation (v). Here we have constructed Ψ_1 in equation (v)

Do you know that!

The celebrated Schrödinger wave equation in terms of **Cartesian** co-ordinates is

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - P) \Psi = 0$$

Here E = Total energy of system P = Potential energy

Ψ = Wave function

x, y, z are Cartesian co-ordinates

If we substitute equation (v) for Ψ_1 in Schrödinger wave equation (vi) then the value of energy lowering will be calculated by

$$E = \frac{\int \Psi_1 H \Psi_1 d\tau}{\int \Psi_1^2 d\tau} \quad \text{Since } \Psi_1 = \Psi_a(1) \Psi_b(2)$$

$$\text{So, } E = \frac{\int \Psi_a(1) \Psi_b(2) H \Psi_a(1) \Psi_b(2) d\tau}{\int [\Psi_a(1) \Psi_b(2)]^2 d\tau} \quad (vii)$$

The energy changes as a function of distance r_{ab} may be shown graphically in Fig. (3.3) (curve I). It should be remembered that electrons '1' and '2' have **not been allowed to exchange**. They are satisfying their own nuclei even coming close to make a bond.

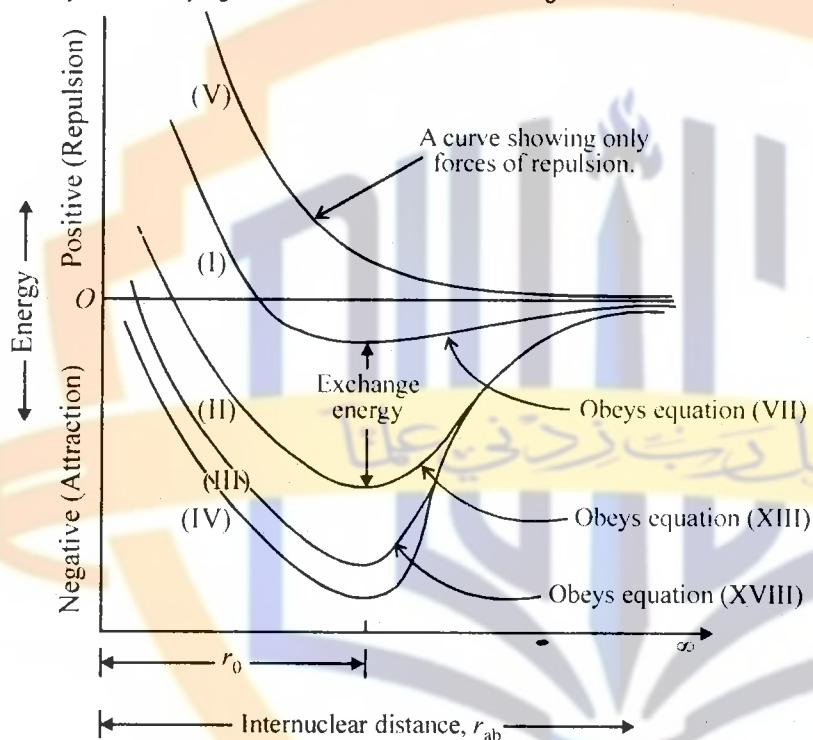


Fig. (3.3): Potential energy curves for hydrogen molecule.

It is a convention that the total energy of the two isolated hydrogen atoms is taken as **zero**. As the two atoms approach one another without exchange of electrons, the electron 1 of H_a is attracted by the nucleus of H_b and electron 2 of H_b is similarly attracted to the nucleus of H_a . As a result there is weak attraction between H_a and H_b . Consequently energy of the system starts decreasing.

With decrease in distance between the two atoms, forces of attraction increase rapidly. Ultimately an **equilibrium internuclear distance (r_0)** is reached. At this distance the system has lowest energy. At distance closer than this, stronger repulsion due to close proximity between the two nuclei are developed. Consequently the energy of the system becomes positive.

The minima in curve (I) represents that a molecule is formed but the bond energy represented by wave function Ψ_1 is about **25.2 kJ mol⁻¹**. This energy is being calculated from equation (vii). When this energy is compared with the experimental value of **431.4 kJ mol⁻¹** for H_2 , then, **the proposed wave function (Ψ_1) seems to be faulty (نقص وال)**. Certain **improvements** are, therefore, necessary to match the **theoretical** and **observed** values. Here **25.2 kJ is theoretical** and **431.4 kJ is experimental** bond dissociation energy. By using first proposed wave function this difference of values i.e., $(431.1 - 25.2) = 405.9 \text{ kJ mol}^{-1}$, is very high. It means that the wave function in equation (v) is **highly approximate**.

(1) **First Improvement:**

When we look at the wave function Ψ_1 , it reveals that electron 1 is associated with atom H_a and electron 2 is associated with atom H_b . When a molecule is formed the two atoms come close together. Their atomic orbitals overlap and we are no longer definite that the atoms retain their individual electrons. We cannot trace any one electron in a system according to **Uncertainty principle**. It clearly states that we cannot exactly locate an electron but only the probability of finding the electron is given. Hence the system of two hydrogen atoms may as well be represented by a **new wave function of two of hydrogens atoms**. The electron 2 is revolving around the nucleus of atom H_a and electron '1' is revolving around the nucleus of H_b . In other words the electrons have got exchanged.

The wave function when exchange of electrons takes place for this may be represented as

$$\Psi_{II} = \Psi_a(2) \cdot \Psi_b(1) \quad (\text{viii})$$

Actually both wave functions Ψ_I of equation (v) and Ψ_{II} of equation (viii) are equally probable representations. The real wave function (Ψ) is likely to be a linear combination of these two wave functions.

$$\Psi = C_1 \Psi_I + C_2 \Psi_{II} \quad (\text{ix})$$

$$\text{or} \quad \Psi = C_1 \Psi_a(1) \Psi_b(2) + C_2 \Psi_a(2) \Psi_b(1) \quad (\text{x})$$

Since the wave function shown by equation (v), (vii) are equally probable, the contribution of wave functions, Ψ_I and Ψ_{II} to the true wave function (Ψ) must be equal. As stated earlier in theorem II, the weight of each wave function is **proportional** to the squares of its coefficients. So,

$$C_1^2 = C_2^2 \text{ or } C_1 = \pm C_2$$

and if for simplicity we assume that $C_1 = 1$, then $C_2 = \pm 1$.

Equation (x) may be represented by two equations

$$\Psi_{\oplus} = \Psi_a(1) \Psi_b(2) + \Psi_a(2) \Psi_b(1) \quad (\text{xi})$$

$$\Psi_{\ominus} = \Psi_a(1) \Psi_b(2) - \Psi_a(2) \Psi_b(1) \quad (\text{xii})$$

Equation (xi) is **second trial wave function** and has two parts. The first part shows when electrons are not exchanged. The second part shows when electrons are exchanged.

Equation (xi) represented by $\Psi_{\oplus}$ is a **symmetric combination**. The reason is that there is no change of sign on exchange of electrons. The energy of the system calculated as a function of internuclear distance, r_{ab} by using wave function Ψ of equation (xi), results in a significant lowering of energy. This can be calculated by putting equation (xi) in equation (vi).

$$E = \frac{\int [\Psi_a(1) \Psi_b(2) + \Psi_a(2) \Psi_b(1)] H [\Psi_a(1) \Psi_b(2) + \Psi_a(2) \Psi_b(1)] d\tau}{\int [\Psi_a(1) \Psi_b(2) + \Psi_a(2) \Psi_b(1)]^2 d\tau} \quad (\text{xiii})$$

It is represented by curve (II).

Concept of Exchange Energy:

The bonding energy is shown in curve (II) and is approximately $302.4 \text{ kJ mol}^{-1}$ and compares well with the experimental curve (IV) which is $431.4 \text{ kJ mol}^{-1}$. The first lowering is 25.2 kJ mol^{-1} . The additional lowering in energy is $302.4 - 25.2 = 277.2 \text{ kJ mol}^{-1}$. This lowering has been obtained by virtue of exchange of electrons between the two hydrogen atoms and is known as **exchange energy** of H_2 molecule.

If we plot the energy of the system using $\Psi_{\odot}$ of equation (xii) as function of distance r_{ab} we obtain curve (V) which shows that the energy curve represented by $\Psi_{\odot}$ is an **anti-symmetric combination**, since there is a change of sign on electron interchange. In this case the two atoms approach each other with their **electrons having parallel spins**. It is clear that $\Psi_{\oplus}$ is the wave function which represents the H_2 molecule. In other words, electrons approach each other with opposite spins. $\Psi_{\oplus}$ is also referred to as Ψ_{cov} as each nucleus is always in possession of one electron at all times.

(2) Second Improvement: Shielding Effect.

The wave function can be improved by considering the **screening effect** of electrons. The forces of attraction between electron 1 and nuclei of H_a and H_b are being lessened due to electron 2. Similarly is the case for electron 2 for nuclei of H_a and H_b . That wave function will be the **third trial** wave function. Since electrons shield one another by varying extents from the nuclear charge, so we must use, Z_{eff} (effective nuclear charge) rather than Z . When Z_{eff} is used, bonding energy of $339.4 \text{ kJ mol}^{-1}$ is obtained. In this way there is improvement of $339.4 - 302.4 = 37 \text{ kJ mol}^{-1}$. This shows that **shielding effect** is a **considerable parameter**.

(3) Third Improvement: Ionic Character.

By the use of wave function $\Psi_{\oplus}$ (or Ψ_{cov}) of equation (xii), we have made a significant improvement in the choice of our wave function. But so far we have assumed that one electron is always associated with one atom. Such a situation may not always be true. **There is the possibility that both electrons 1 and 2 are either on atom H_a or H_b .** The wave function describing such a state would be a combination of equations (xiv) and (xv).

Its wave function may be written as,

$$\Psi_{\text{III}} = \Psi_a(1) \Psi_a(2) \quad (\text{xiv})$$

This is when both electrons are close to atom H_a . When both electrons are close to atom 'b' then a new wave function Ψ_{IV} equation (xv) is,

$$\Psi_{\text{IV}} = \Psi_b(1) \Psi_b(2) \quad (\text{xv})$$

The resulting wave function representing both electrons on one nucleus may be summed by taking equations (xiv) and (xv).

$$\Psi_{\text{ion}} = \Psi_a(1) \Psi_a(2) + \Psi_b(1) \Psi_b(2) \quad (\text{xvi})$$

It is called Ψ_{ion} . Since it represents a state in which ionic character of the bond forms. It also contributes to the structure of H_2 molecule. If we take function Ψ_{cov} together with Ψ_{ionic} , then the real wave function may be written as

$$\Psi = \Psi_{cov} + \lambda \Psi_{ion} \quad (xvii)$$

$$\text{or } \Psi = \underbrace{\Psi_a(1) \Psi_b(2) + \Psi_a(2) \Psi_b(1)}_{\text{\{from equation (x)\}}} + \lambda \underbrace{[\Psi_a(1) \Psi_a(2) + \Psi_b(1) \Psi_b(2)]}_{\text{\{from equation (xiv)\}}} \quad (xviii)$$

Here λ in equation (xviii) represents the **mixing coefficient**. It tells us how much Ψ_{ion} wave function is to be introduced in the real wave function Ψ of equation (xviii). This equation (xviii) is also called **fourth trial wave function**. By applying the **variation method**, we can adjust the value of λ so as to obtain the lowest energy. In this way, we can measure the extent of ionic character in the covalent bond. For H_2 , the value of λ is about **0.17**.

If we now calculate the energy of the system using wave function as given by equation (xviii) we get a new energy curve (III) in Fig. (3.3). The energy associated with curve (III) is **362.4 kJ mol⁻¹** and is an improvement over our earlier results and is quite close to the experimental value of **431.4 kJ mol⁻¹**.

This treatment which we have discussed for the formation of H_2 molecule is essentially the pairing of two electrons, one from each bonded atom. This is called **valence bond method**. The valence bond method is also referred to as method of **trial electron pairing**. Here we construct trial wave functions. It is obvious even from equation (xi) that the atoms must approach one another with the spins of their electrons in opposite directions. It follows that the formation of covalent bond is explained by the lowering of the potential energy of the electrons, resulting from the increased electron density in the internuclear space.

Trial Wave Functions:

All those equations which are developed under the proposal of various wave functions are called **trial wave functions**. They are also called **approximate wave functions**. When we further go on constructing better and better wave functions, then ultimately we reach E as **-431.2 kJ mol⁻¹** and bond distance of **0.74 Å**. The rest part of this theory will be discussed after **VSEPR theory**.

3.3 Valence Shell Electron Pair Repulsion (VSEPR) Theory

VSEPR theory is the simplest theory for predicting the geometries of molecules. This theory was proposed by **Sidgwick and Powell in 1940** and was given the final shape by **Gillespie and Nyholm (1957)**. It is based on the repulsive interactions between the electron pairs in the valence shell.

Main Points of VSEPR Theory:

1. **Geometry** of a molecule depends upon the total number of valence shell electron pairs (bonded or non-bonded) around the central atom.
2. In order to minimise electron-electron repulsion or to acquire a state of minimum energy or maximum stability, the valence shell electron pairs keep **as far apart as possible**.
3. If more than one geometrical arrangement is possible for a given number of bonded and non-bonded pairs, then the most stable arrangement is in accordance with two rules.
 - (a) Lone pair-lone pair (*lp-lp*) repulsion is greater than lone pair-bond pair (*lp-bp*) repulsion which in turn is greater than bond pair-bond pair (*bp-bp*) repulsion. That is

$$lp-lp > lp-bp > bp-bp$$

- (b) Repulsion between electron pairs at vertices greater than 115° apart can be neglected.

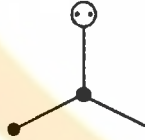
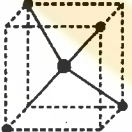
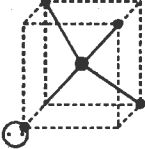
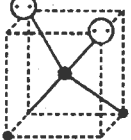
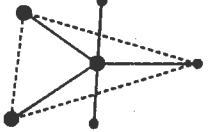
In the formation of a bond, the central atom shares its valence electrons with the surrounding atoms.

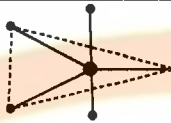
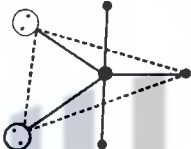
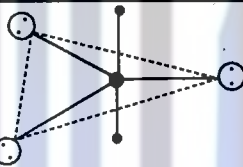
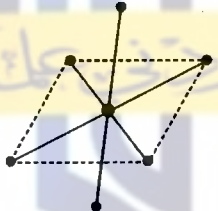
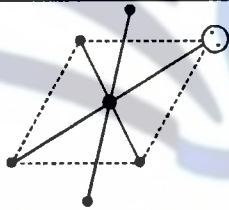
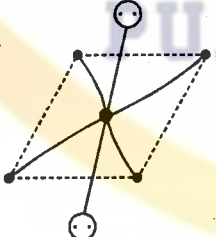
Concept of Bond Angle:

The bond angle is the angle formed by two surrounding atoms with the central atom at the vertex. Ideal angles are observed when all the bonds around a central atom are identical and are connected to the same type of atom. When this is not the case, such as when multiple bonds, lone pairs, or different surrounding atoms are present, the bond angles deviate from the ideal angles.

We use the **VSEPR model** to explain the molecular shapes, bond angles and bond lengths. We observe them through various types of spectroscopy. VSEPR predictions are in accordance with our observations.

Table (3.1): VSEPR Theory and Geometry of Molecules

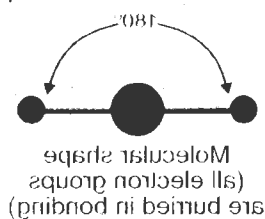
Total number of electron pairs around the central atom	Number of lone pairs	Molecular shape	Example
2	0	 Linear	BeF_2 , CO_2
3	0	 Trigonal planar	SO_3 , BF_3
3	1	 Angular	:SO_2
4	0	 Tetrahedral	CH_4 , CF_4 , SO_4^{2-}
4	1	 Trigonal pyramidal	NH_3 , PF_3
4	2	 Angular	$\text{H}_2\ddot{\text{O}}$, $\text{H}_2\ddot{\text{S}}$
5	0	 Trigonal bipyramid	PF_5 , AsF_5

5	1		See-saw planar :SF ₄
5	2		T-shaped :ClF ₃
5	3		See-saw planar :XeF ₄ , I ₃ [⊖]
6	0		Octahedral SF ₆ , PF ₆ [⊖]
6	1		Square pyramid :IF ₄ , BrF ₅
6	2		Square planar XeF ₄ , IF ₄ [⊖]

3.3.1 Molecular Shape with Two Electron Groups (Linear Arrangement)

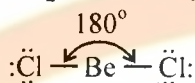
AX₂ Type Compounds.

When two electron groups repel each other, they lie on opposite sides of the central atom in a straight line. The **linear arrangement** of electron groups results in a linear molecular shape and a bond angle of 180°.

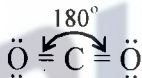


BeCl₂:

Gaseous beryllium chloride (BeCl₂) is an example of a linear molecule (AX₂).

**CO₂:**

In carbon dioxide, the central carbon is surrounded by two double bonds:



Each double bond acts as a separate electron group and stays 180° away from the other, so CO₂ is linear. Similarly CS₂, HCN, BeF₂ etc. have linear structure.

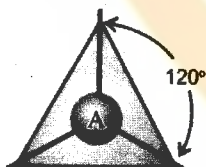
Remember that!

The lone pairs on the O atoms of CO₂ or on the Cl atoms of BeCl₂ are not involved in the molecular shape. Only the valence electron groups around the central atom influence the molecular shapes.

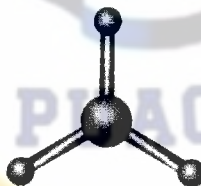
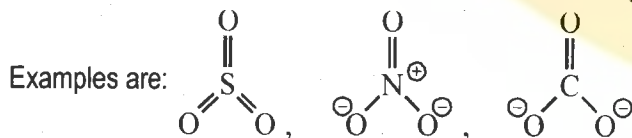
3.3.2 Molecular Shapes with three Electron Groups (Trigonal Planar Arrangement)

(1) AX₃ Type Compounds:

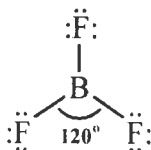
Three electron groups around the central atom repel each other. They lie at the corners of an **equilateral triangle**. This is the **trigonal planar arrangement**, and the ideal bond angle is 120°.



TRIGONAL PLANAR

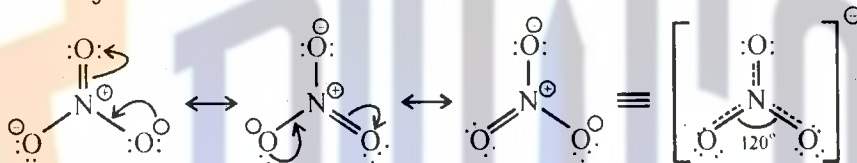
Trigonal planar
AX₃ type**Structure of BF₃:**

When the three electron groups are bonding groups, the name of the molecular shape is also **trigonal planar**. Boron trifluoride (BF₃) has **six electrons** around the central B in three single bonds to F atoms. The molecule is **flat** with all four atoms lying in a plane.



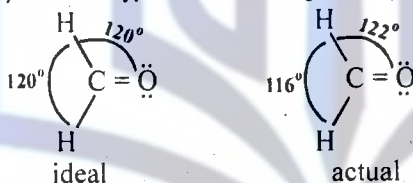
Resonance in NO_3^- and CO_3^{2-} Ions:

The nitrate (NO_3^-) and carbonate (CO_3^{2-}) are polyatomic ions. They have **trigonal planar** shapes. Each of them shows three **resonance structures**. The resonance hybrid has three identical $\frac{1}{3}$ bonds, so the ideal bond angle of 120° is observed: Look at three resonance structures of NO_3^- ion.



Structure of Formaldehyde (HCHO):

When the surrounding atoms or the electron groups are not identical, **non-ideal bond angles** occur. Consider formaldehyde (CH_2O). Its trigonal planar shape is due to two types of surrounding atoms (O and H) and two types of electron groups (single and double bonds):

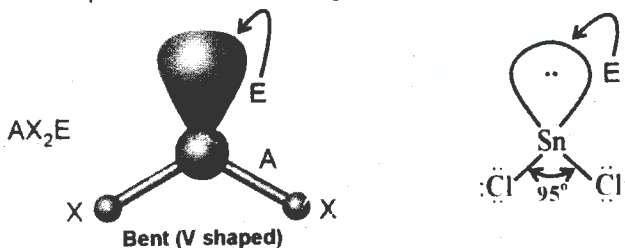


The actual bond angles differ from the ideal ones. The reason is that the **four electrons** in the double bond repel the **two electrons** in each of the two single bonds more strongly than they repel each other. Four electrons of a double bond occupy more space.

(2) AX_2E Type Compounds:

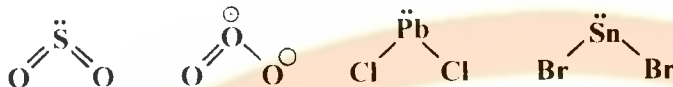
Positions of atoms define a **shape**, so when one of the three electron groups is a lone pair represented by E in (AX_2E), the molecule is **bent** or **V shaped**. Here E represents the lone pair.

The shape of **gaseous tin (II) chloride (SnCl_2)** has the three electron groups in a trigonal plane, with the lone pair at one of the triangle's corners.



Since a lone pair is held by only one nucleus, it exerts a stronger repulsion than a bonding pair. Thus, a lone pair repels bonding pairs more strongly than bonding pairs repel each other. This repulsion increases the angle between lone pair and bonding pair, which decreases the angle between bonding pairs. Note the large decrease from the ideal 120° to 95° angle in SnCl_2 :

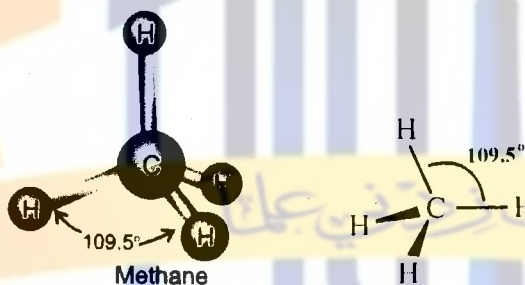
Some other examples in this respect are as follows:



3.3.3 Molecular Shapes with Four Electron Groups (Tetrahedral Arrangement)

(1) AX_4 Type Compounds. (Structure of CH_4)

All molecules or ions with four electron groups around a central atom, adopt the tetrahedral arrangement. Perspective drawings for CH_4 is as shown.



(The effect of bond-pair repulsions on bond angle of a perfect tetrahedral.)

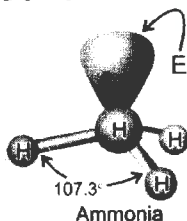
The normal lines represent electron groups in the plane of the page, one **wedge** is the bond of a group lying **towards** you above the page, and the other is the bond of a group lying **away** from you below the page.

When the four electron groups are involved in bonding, the molecular shape is called tetrahedral (AX_4). This is an extremely common geometry in organic molecules.

The molecules of IV-A group as ${}_6\text{C}$, ${}_{14}\text{Si}$, ${}_{32}\text{Ge}$, ${}_{50}\text{Sn}$ and ${}_{82}\text{Pb}$ give such structures as SiH_4 , SiX_4 , GeX_4 , SnH_4 , SnX_4 , PbCl_4 etc.

(2) AX_3E Type Compounds (Structure of NH_3)

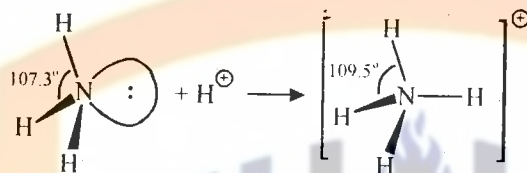
When one of the four electron groups in the tetrahedral arrangement is a lone pair, the molecular shape is that of a **trigonal pyramid** (AX_3E). This is a tetrahedron with one point "missing". As expected, the measured bond angle is slightly less than the ideal 109.5° . In ammonia (NH_3), for example, the lone pairs forces the $\text{N}-\text{H}$ bonding pairs together, and the $\text{H}-\text{N}-\text{H}$ bond angle is 107.3° instead of 109.5° .



Some other examples are PH_3 , AsH_3 , SbH_3 , BiH_3 , PCl_3 , NF_3 , AsCl_3 etc.

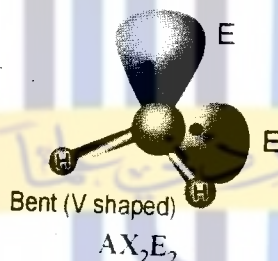
When NH_3 and H^+ form an ammonium ion, for example, the lone pair on trigonal pyramidal NH_3 forms a covalent bond coordinated to H^+ to form **tetrahedral** NH_4^+ . In NH_4^+ the $\text{H}-\text{N}-\text{H}$ bond angle expands. The reason is that the repulsions due to the lone pair decrease,

because another bonding pair is developed. The status of four electron pairs is equal and all bond angles in NH_4^+ are 109.5° each.



(3) **AX_2E_2 Type Compounds [Structure of (H_2O)]**

When the four electron groups around the central atom include two bonding and two nonbonding groups, the molecular shape is **bent** or **V shaped** (AX_2E_2).

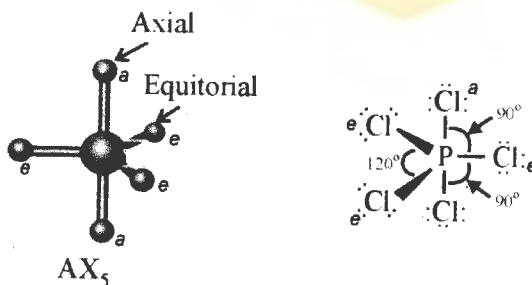


Some other compounds are $\text{H}_2\ddot{\text{S}}$, $\ddot{\text{O}}\text{F}_2$, $\ddot{\text{S}}\text{Cl}_2$ etc.

3.3.4 Molecular Shapes with Five Electron Groups (Trigonal Bipyramidal Arrangement)

(1) **AX_5 Type Compounds: [Structure of PCl_5]**

When five electron groups maximize their separation, they form the trigonal bipyramidal arrangement, *i.e.*, two trigonal pyramids sharing the same base. Note that there are two positions for surrounding electron groups and two ideal bond angles. Three **equatorial groups** lie in a trigonal plane that includes the central atom. Two **axial groups** lie above and below this plane. Therefore, 120° bond angles separate equatorial groups, and a 90° angle separate axials from equatorials. In general, the greater the bond angle, the weaker the repulsions, so **equatorial-equatorial** (120°) repulsions are weaker than **axial-equatorial** (90°) repulsions. The structure of PCl_5 is given below.

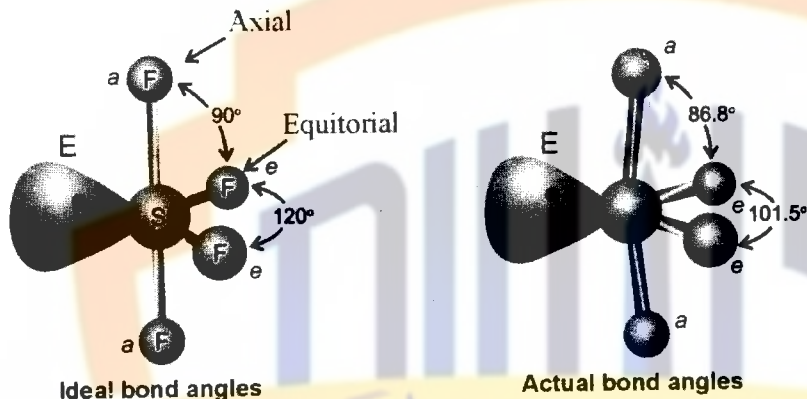


There are no lone pairs on P and As in PCl_5 and AsF_5 .

(2) **AX_4E Type, Compounds [Structure of Sulphur Tetrafluoride ($\ddot{\text{S}}\text{F}_4$)]**

Lone pairs exert stronger repulsive forces than bonding pairs, so to minimize repulsions, the lone pairs occupy equatorial positions. With one lone pair present at an equatorial po-

sition, the molecule has a **seesaw shape** (AX_4E). Sulphur tetrafluoride (SF_4), has this shape. Note the ideal bond angles and actual bond angles. The equatorial lone pair repels all four bonding pairs to lower the actual bond angles.



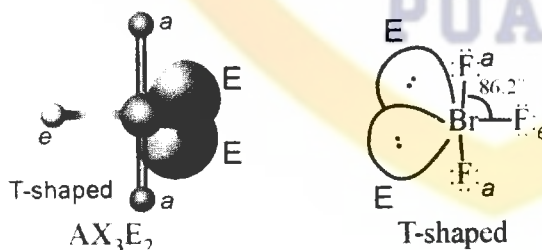
Ideal and actual bond angles in SF_4 .

To minimize 90° repulsions, the single lone pair in SF_4 occupies an equatorial position of the trigonal bipyramid. The lower actual bond angles result from lone-pair repulsions of the bonding pairs.

Some other examples are $2SCl_2F_2$, $IF_4^\ominus$, $IO_2F_2^\ominus$.

(3) AX_3E_2 Type Compounds [Structure of Bromine Trifluoride (BrF_3)]

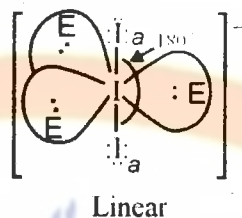
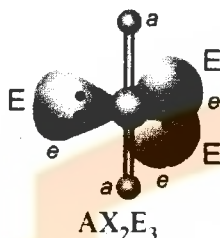
The tendency of lone pairs to occupy equatorial positions causes molecules with three bonding groups and two lone pairs to have a T shape. Bromine trifluoride (BrF_3), has this shape and the predicted angles decrease from the ideal 90° $F-Br-F$ bond angle: An other example is chlorine trifluoride ClF_3 . (e for equatorial and a for axial)



The molecular shapes of the trigonal bipyramidal electron-group arrangement.

(4) AX_2E_3 Type, Compounds [Structure of Iodate Ion ($I_3^\ominus$)]

Molecules with two bonding groups have three lone pairs in equatorial positions. The two bonds are in axial positions, giving the molecule a linear shape. For example, the tri-iodide ion ($I_3^\ominus$), which forms, when I_2 dissolves in aqueous $I^\ominus$ solution is linear:



An other example is $\ddot{Xe}F_2$. Xe has three lone pairs and molecule is linear.

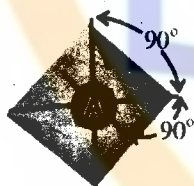
3.3.5 Molecular Shapes with Six Electron Groups (Octahedral Arrangement)

AX_6 Type Compounds:

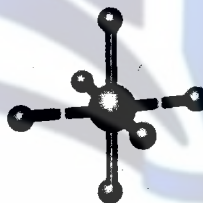
Six electron groups point to the corners of an **octahedron**. It is an eight-sided regular **polygon** with six equidistant **vertices**. The six positions are equivalent. So all the groups have a 90° ideal bond angle. Three important molecular shapes are within this arrangement.

Sulphur Hexafluoride (SF_6):

With six bonding groups, the molecular shape is **octahedral** (AX_6). When seesaw-shaped SF_4 , for example, reacts with additional F_2 , the central S-atom expands its valence shell further to form sulphur hexafluoride (SF_6):



Octahedral
(AX_6)

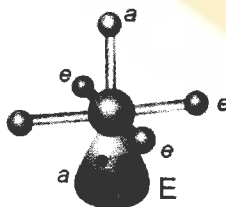


Octahedral
(AX_6)

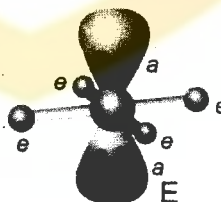


SF_6

Similarly, the compounds of AX_5E having one lone pair and five bond pairs are $\ddot{Br}F_5$, $\ddot{Te}F_4$ and $\ddot{Xe}OF_4$.



Square pyramidal (AX_5E)



Square planar (AX_4E_2)

The molecular shapes of the octahedral electron-group arrangement.

Some other compounds of type AX_4E_2 are $\ddot{Xe}F_4$ and ICl_4^- .

Limitations of VSEPR Theory:

- (i) VSEPR theory is applicable to **covalent compounds only** and cannot be applied to **ionic compounds**.
- (ii) This theory does not make any **distinction** between **s, p, d and f orbitals**.

- (iii) The theory does not help in the prediction of shapes of coordination compounds.
- (iv) VSEPR theory simply predicts that molecules such as NH_3 , $\text{H}_2\ddot{\text{O}}$, CHCl_3 , etc. are distorted from the regular geometrical shapes but does not quantify the extent of distortion.

3.4 Valence Bond Theory (Hybridization of Orbitals)

The mixing of atomic orbitals to give equal number of symmetrically hybridized orbitals is referred to as hybridization.

Rule for Hybridization

Some of the rules for orbital hybridization are as follows:

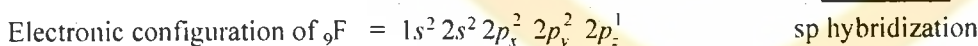
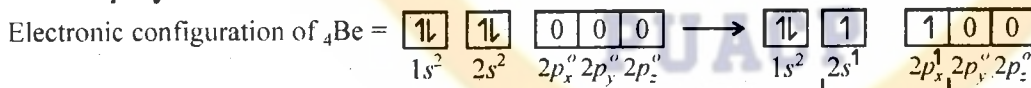
- (1) Number of hybrid orbitals produced, is equal to the number of orbitals undergoing hybridization.
- (2) The type of hybridization can predict the geometry and bond angles of a molecule.
- (3) That orbital which has been used to build up a hybrid orbital becomes no longer available to hold electron in its pure form.
- (4) Orbitals of similar energies belonging to the same atom or ion can hybridize together.

3.4.1 Structure of BeF_2

Let us discuss the structure of BeF_2 molecule. One of electrons of $2s$ is promoted to say $2p_x$ orbital.

$2s^1$ and $2p_x^1$ hybridize to give two sp hybridized orbitals. An sp orbital has 50% s character and 50% p character.

sp -Hybridization:



These two hybridized orbitals overlap separately with the p -orbitals of fluorine having one electron each to form two Be-F bonds. Shape of BeF_2 molecule can be represented as shown in Fig. (3.4). There are two bonded electron pairs. Therefore molecule is linear with two orbitals at an angle of 180° to each other. The overlap for σ -bond is due to head on overlapping of sp of Be and p_z of F-atom.

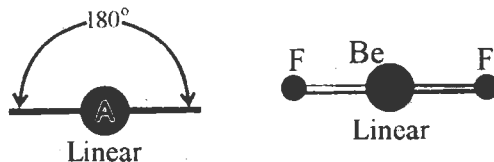


Fig. (3.4): Molecular shape of the linear electron-group arrangement.

3.4.2 sp^2 Hybridization or Trigonal Hybridization

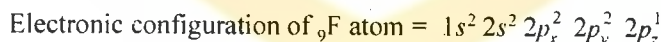
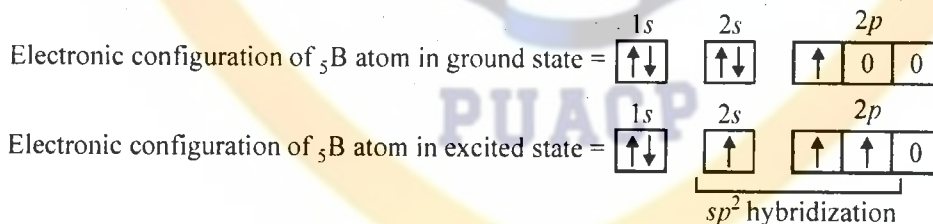
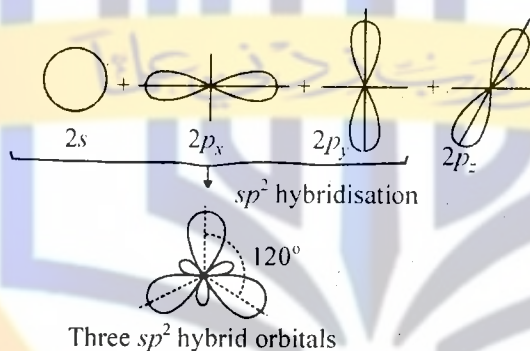
When one s and two p -orbitals combine together to produce equivalent type of orbitals, sp^2 hybridization takes place.

(1) Structure of BF_3 :

Let us discuss the structure of BF_3 . Electronic configuration of boron in ground state is $1s^2 2s^2 2p_x^1$. One of the two $2s$ electrons gets promoted to next available vacant orbital say $2p_y$. The electronic configuration of boron in the excited state is $1s^2 2s^1 2p_x^1 2p_y^1 2p_z^0$. (Remember that promoted electron of $2s$ can go to any of the $2p$ orbital i.e., $2p_y$ or $2p_z$).

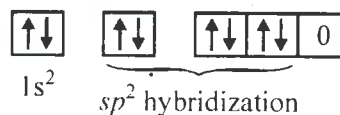
Now, the electrons belong to two different types of orbitals i.e., one is $2s$, and two are $2p$. To get equivalent bonds these three orbitals hybridize to form three new equivalent hybridised orbitals. In this case one $2s$ and two $2p$ orbitals are involved to form three hybridised orbitals. This type of hybridization is, therefore, known as sp^2 hybridization or trigonal hybridization. It has 33.3% s and 66.6% p -character.

These three orbitals lie in **one plane** and are directed towards the **three corners of an equilateral triangle**, so that the angles between them are 120° .



Three F-atoms share three electrons of their $2p_z$ orbital with electrons of $2p^2$ hybridized orbitals.

Configuration of B atom after making bonds with three F-atoms.



Boron atom is capable of forming three bonds by overlapping with, three half-filled orbitals of other atoms. The overlap is between sp^2 of B-atom and p -orbitals of F atoms.

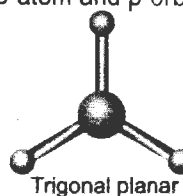
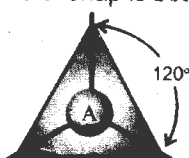


Fig. (3.5): The two molecular shapes of the trigonal planar electron-group arrangement.

Shape of BF_3 molecule can be represented as shown in Fig. (3.5). The other examples are SO_3 , BCl_3 , NO_3^- , CO_3^{2-} etc.

(2) Structure of Stannous Chloride (SnCl_2):

In stannous chloride the tin ($_{50}\text{Sn}$) atom has electronic configuration as

Electronic configuration of $_{50}\text{Sn}$ atom in ground state = $5s \uparrow\downarrow$ $5p \uparrow\uparrow 0$ $5d 0 0 0 0 0$

In order to make SnCl_2 there is no need for the promotion of electrons and three orbitals hybridize as such with 4-electrons in them.

= $5s \uparrow\downarrow$ $5p \uparrow\uparrow 0$ $5d 0 0 0 0 0$
 sp^2 hybridisation

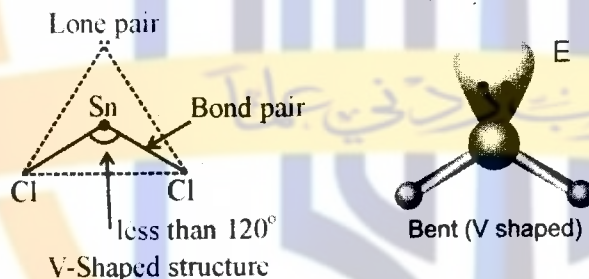
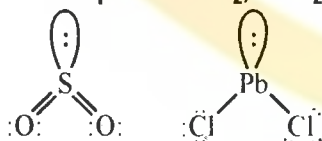


Fig. (3.6): Formation of stannous chloride.

In SnCl_2 , two unpaired electrons in outermost shell of tin (Sn) are used for bond formation with two atoms of chlorine. One electron pair is left unused as lone pair. Thus there are three electron pairs, two bond pairs and a lone pair. According to its hybridization, it is expected to be trigonal planar Fig. (3.6). One of the three positions is occupied by lone pair of electrons. The expected angle is 120° . But the actual angle is less than 120° . This is because of a greater repulsion between the bond pair (bp) and lone pair (lp) than between the bond pair (bp) and bond pair (bp). The other examples are SO_2 , PbCl_2 etc.



The bonds are produced in SnCl_2 and PbCl_2 by head on overlap of p orbitals of Cl-atoms and sp^2 of Sn and Pb.

3.4.3 sp^3 Hybridization or Tetrahedral Hybridization

When one s orbital and three p orbitals combine together to produce equivalent type of orbitals, sp^3 hybridization takes place.

(1) Structure of CH_4 :

Four sp^3 hybridized orbitals are produced which have the same energy and shape and are oriented at an angle of $109^\circ 28'$ to one another Fig (3.7). This is also known as **tetrahedral hybridization**, because the hybridized orbitals are oriented towards the four corners of

a tetrahedron. The nucleus of carbon atom occupies the centre of tetrahedron. In this position, hybridized orbitals experience **minimum repulsive forces**.

Electronic configuration of carbon ($_6\text{C}$) in ground state = $\boxed{1\downarrow} \quad \boxed{1\downarrow} \quad \boxed{1 \quad 1 \quad 0}$
 $1s \quad 2s \quad 2p_x \quad 2p_y \quad 2p_z$

And carbon ($_6\text{C}$) in excited state = $\boxed{1\downarrow} \quad \boxed{1} \quad \boxed{1 \quad 1 \quad 1}$
 $1s \quad 2s \quad 2p_x \quad 2p_y \quad 2p_z$
 sp^3 hybridization

There are four orbitals having unpaired electrons. These $2s$, $2p_x$, $2p_y$ and $2p_z$ orbitals will first of all hybridize. Four sp^3 hybridized orbitals produced are arranged tetrahedrally and are shown in Fig. (3.7).

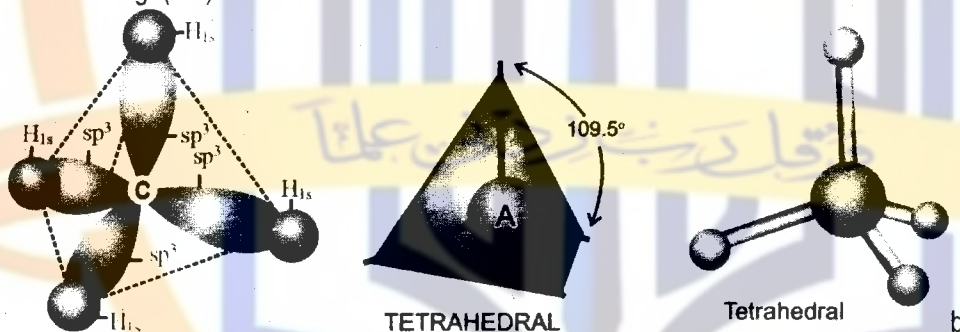


Fig. (3.7): The molecular shapes of the tetrahedral electron-group arrangement.

All the four hybridized orbitals overlap with the $1s$ orbitals of four hydrogen atoms. All these overlappings take place along the internuclear axes. Hence, four carbon-hydrogen σ bonds are formed. These bonds are due to $s-sp^3$ overlapping.

Other examples are CCl_4 , CBr_4 , SiH_4 , SiCl_4 , GeH_4 , GeCl_4 , SnCl_4 , PbCl_4 etc.

(2) Structure of Ammonia (NH_3):

Electronic configuration of nitrogen ($_7\text{N}$) in ground state = $\boxed{1\downarrow} \quad \boxed{1\downarrow} \quad \boxed{1 \quad 1 \quad 1}$
 $1s^2 \quad 2s^2 \quad 2p_x \quad 2p_y \quad 2p_z$
 sp^3 hybridization

Here, there is no possibility of shifting of electron from $2s$ level to $2p$ level as there is no empty $2p$ orbital. $2s$ orbital, although incapable of overlapping participates in hybridization. This will give rise to sp^3 hybridization.

Hybridized orbitals are oriented towards the corners of a tetrahedron. Three sp^3 hybridized orbitals overlap with the $1s$ orbitals of three hydrogen atoms. The fourth corner is occupied by lone pair. We expect an angle of $109^\circ 28'$ (109.5°) between $\text{H} - \text{N} - \text{H}$ in tetrahedral geometry. **Actually this angle is 107.5° .** This contraction (انكسار) in angle takes place because the lone pair of electrons repels the shared pair of electrons or bonded pair of electrons with a greater force, thereby reducing the angle Fig (3.8).

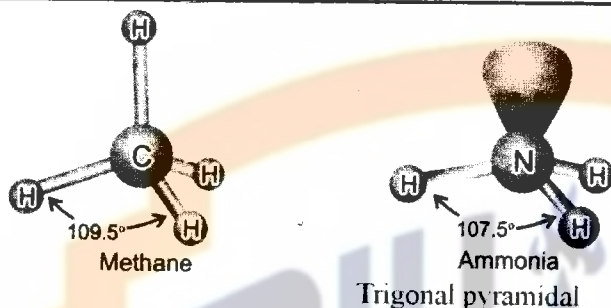


Fig. (3.8): The effect of lone-pair repulsions on bond angle in NH_3 in comparison to CH_4 .

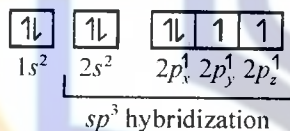
Similarly we can justify the shapes of PH_3 , SbH_3 and BiH_3 . PCl_3 , PBr_3 , SbCl_3 , BiCl_3 , ClO_3^- , H_3O^+ .

(3) Structure of Water ($\text{H}_2\ddot{\text{O}}$)

Electronic configuration of oxygen (${}_8\text{O}$) atom =

1↓	1↓	1↓	1↓	1↓
$1s^2$	$2s^2$	$2p_x$	$2p_y$	$2p_z$

There is no promotion of electrons, so electronic configuration of ${}_8\text{O}$ is,



These outermost orbitals having six electrons can participate in hybridization. All four orbitals hybridize and two of the positions on the tetrahedron are occupied by lone pairs of electrons. The other two hybridized orbitals will overlap with the $1s$ orbitals of two hydrogen atoms to form a water molecule as shown in Fig. (3.9). We expect an angle of $109^\circ 28'$ in HOH . Actually, the angle is 104.5° . This is due to stronger repulsion of two lone pairs of electrons with shared pair of electrons.

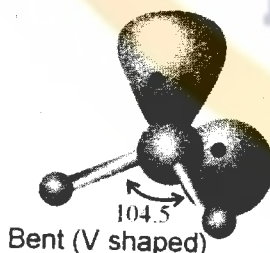


Fig. (3.9): In $\text{H}_2\ddot{\text{O}}$ two lone pairs cause greater repulsions and the bond angle is decreased to 104.5° .

Similarly we can discuss the structures of $\text{H}_2\ddot{\text{S}}$, $\text{H}_2\ddot{\text{Se}}$, $\text{H}_2\ddot{\text{Te}}$, $\ddot{\text{S}}\text{Cl}_2$ etc.

(4) Structure of Hydronium Ion [H_3O^+]

The formation of coordinate covalent bond in H_3O^+ may be represented as follows.

A coordinate covalent bond results when an empty orbital present in the other atom accommodates a lone pair of an other atom. It is important to note that the coordinate covalent bond once formed is indistinguishable from a covalent bond. The H^+ ion has an empty s orbital which overlaps with one of the orbitals of oxygen atom containing a lone pair of electrons forming hydronium ion, H_3O^+ . This is illustrated in Fig. (3.10). This lone pair of oxygen is present in an sp^3 orbital.

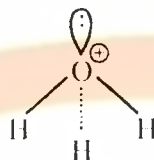


Fig. (3.10): The geometry of H_3O^+ ion. One of the lone pair of O-atom is at the fourth corner of tetrahedron.

Compare $\text{H}_2\ddot{\text{O}}$ and H_3O^+

The shape of $[\text{H}_3\text{O}]^+$ will continue to remain tetrahedral because the hybridization involved remains sp^3 . In H_2O , two tetrahedral positions are occupied by lone pairs of electrons but in H_3O^+ only one tetrahedral position is occupied by lone pair of electrons and this lone pair is still there on the fourth corner of positive oxygen.

3.4.4 sp^3d Hybridization

Mixing of one s, three p and one d atomic orbitals gives five sp^3d hybrid orbitals and process is called sp^3d hybridization.

These five sp^3d orbitals are directed towards the corners of a **trigonal bipyramid** as shown in Fig. (3.11a). They form three coplanar bonds with bond angles of 120° , remaining two bonds are at right angles to this plane one above and other below. Let us take example of PF_5 in this case.

(1) Structure of PF_5 Molecule:

The outer electronic configuration of phosphorus in the ground state is given.

In order to have the requisite number of half-filled orbitals available for hybridization purpose, one of the 3s electrons is made to jump to one of the 3d empty orbitals, thereby giving rise to 5 half-filled orbitals for bonding purposes. These five orbitals hybridize to give five sp^3d orbitals. Now five orbitals are available for combination which permit the formation of five covalent bonds with five fluorine atoms.

Electronic configuration of $_{15}\text{P}$ in ground state = $1s^2 2s^2 2p^6 3s^1 3p_x^1 3p_y^1 3p_z^1 3d^0 3d^0 3d^0 3d^0 3d^0$

Electronic configuration of $_{15}\text{P}$ in excited state = $1s^2 2s^2 2p^6 3s^1 3p_x^1 3p_y^1 3p_z^1 3d^1 3d^0 3d^0 3d^0 3d^0$

$\underbrace{\hspace{10em}}_{sp^3d \text{ hybridization.}}$

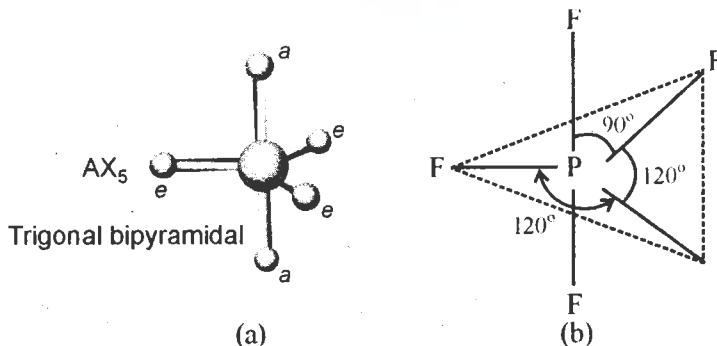


Fig. (3.11): The four molecular shapes of the trigonal bipyramidal electron-group arrangement.

But all the hybridised orbitals are **not equivalent**. Three of these are oriented towards the three corners of an **equilateral triangle** making an angle of **120°** between them. The bonds formed by these orbitals are known as **equatorial bonds**. Remaining two orbitals are oriented **at right angles** to the plane of first set of three orbitals. Thus PF_5 has trigonal bipyramidal shape as shown in Fig. (3.11)

The other molecules as PCl_5 , PBr_5 , AsCl_5 etc. can be similarly explained.

(2) Structure of SF_4 Molecule:

Electronic configuration of sulphur ($_{16}\text{S}$) atom in ground state. = $3s \uparrow\downarrow \quad 3p \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \quad 3d \quad \quad \quad$

Sulphur ($_{16}\text{S}$) atom in excited state. = $3s \uparrow\downarrow \quad 3p \uparrow \uparrow \uparrow \quad 3d \uparrow \quad \quad \quad$

sp^3d hybridization

To have four unpaired electrons, one electron from $3p_x$ orbital is unpaired and promoted to $3d$ orbital. Thus five orbitals are used, four contain shared or bonded pairs of electrons and one contains a lone pair. sp^3d hybridization occurs, giving an expected trigonal bipyramidal shape.

Actually the four fluorine atoms are joined tetrahedrally to sulphur atom in an irregular manner due to presence of a lone pair Fig. (3.12)

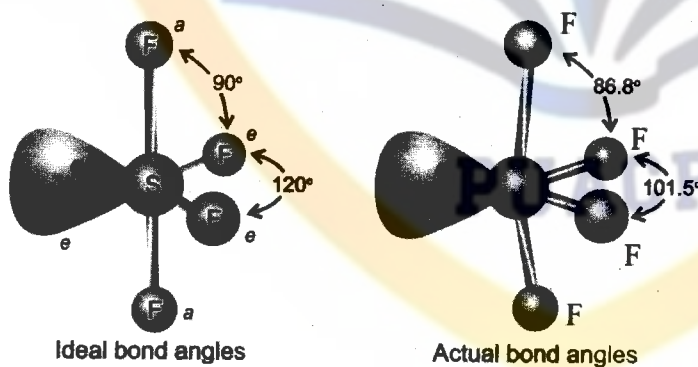


Fig. (3.12): Distorted trigonal bipyramid structure of SF_4 .

(3) Structure of $[\text{ICl}_2]^\ominus$ Ion:

$\text{ICl}_2^\ominus$ ion has been found to be linear in shape. This can be explained by assuming sp^3d hybridization of the orbitals of iodine. sp^3d hybridization leads to trigonal bipyramidal shape. Two chlorine atoms occupy axial (a) positions because the repulsions will be the minimum in this arrangement. **One $\text{Cl}^\ominus$ ion will enter one of the sp^3d hybrid orbitals which is vacant.** It will make a σ -bond. Another sp^3d hybrid orbital containing single electron will overlap with one p -orbital of one chlorine atom to form one bond. Thus, in actual appearance the $\text{ICl}_2^\ominus$ ion will have **linear shape** because the three base positions of the triangular bipyramid will be occupied by lone pairs of electrons as shown in Fig. (3.13)

Ground state electronic configuration of iodine $_{53}\text{I} = \dots 5s^2 5p_x^2 5p_y^2 5p_z^1 5d^0$

Hybridization without excitation $_{53}\text{I} = \boxed{1\downarrow} \boxed{1\downarrow} \boxed{1\downarrow} \boxed{1} \boxed{0}$

$5sp^3d$ orbitals, $\text{Cl}^{\ominus}$ goes here
 $\text{Cl}^{\ominus}$ goes here.

The other examples are XeF_2 , $\text{I}_3^{\ominus}$, $\text{IF}_2^{\ominus}$ etc.

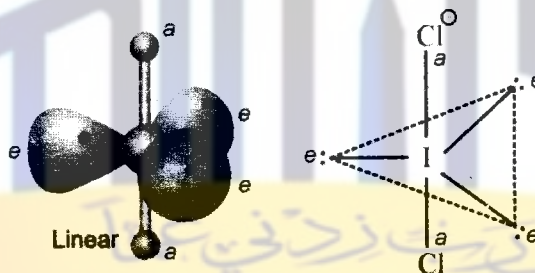


Fig. (3.13): Linear geometry of $\text{ICl}_2^{\ominus}$ ion.

(4) Structure of ClF_3 Molecule:

The outer electronic configuration of chlorine in ground state is $3s^2 3p_x^2 3p_y^2 3p_z^1$.

In the formation of ClF_3 molecule, one of the paired electrons in $3p$ orbital is promoted to $3d$ orbital.

The promotion of $3p$ electron is followed by hybridization involving one s , three p and one d orbitals to give sp^3d hybridization. The chlorine atom is therefore surrounded by five orbitals, three containing a pair of electrons while the other two contain lone pair of electrons.

Electronic configuration of chlorine ($_{17}\text{Cl}$) atom in ground state.



Electronic configuration of chlorine ($_{17}\text{Cl}$) atom in excited state.



sp^3d hybridization

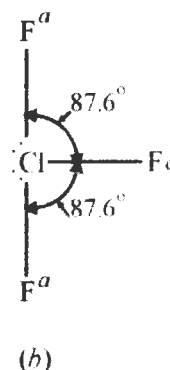
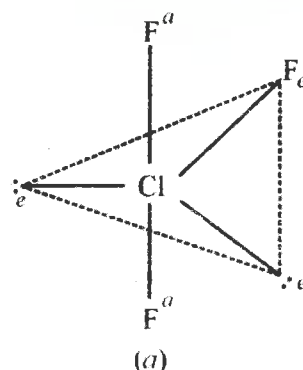
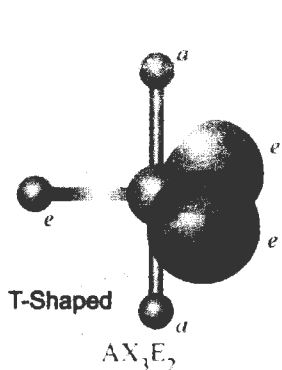


Fig. (3.14): Structure of ClF_3 molecule.

As two positions of the trigonal bipyramid are occupied by lone pairs, the molecule has a T-shape. The angle F — Cl — F expected from this geometry is 90° Fig. (3.14). However, actual angle is 87.6° . This is due to repulsion from lone pairs which result in **contraction of angle**. The other examples are $\ddot{\text{Cl}}\text{F}_3$ and $\ddot{\text{Br}}\text{F}_3$ Fig. (3.14).

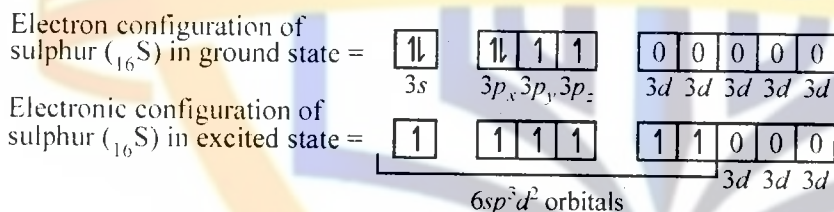
3.4.5 sp^3d^2 Hybridization

A combination of one s, three p and two d atomic orbitals gives six equivalent sp^3d^2 hybrid orbitals directed octahedrally.

This type of combination is known as sp^3d^2 hybridization. Hybrid orbitals are oriented towards the corners of an **octahedron**. Let us discuss a few examples.

(1) Structure of Sulphur Hexafluoride (SF_6):

Outer electronic configuration of sulphur atom in the ground state is $3s^2 3p_x^1 3p_y^1 3p_z^1$. To get the requisite number of half-filled orbitals for hybridization purposes, one of the 3s and one of $3p_x$ electrons are made to jump to two of the 3d empty orbitals. Now six unpaired electrons are available in six orbitals.



The six hybrid orbitals are now directed towards the corners of a regular **octahedron**. These six hybrid orbitals overlap with the p-orbitals of six fluorine atoms. So SF_6 molecule has an **octahedral structure** as shown in Fig. (3.15). The other example is IOF_5 .

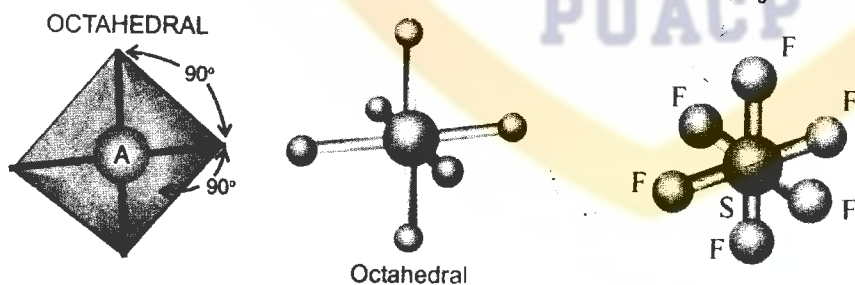
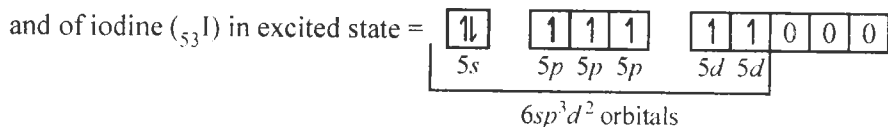
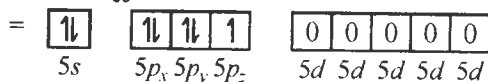


Fig. (3.15): The molecular shapes of the octahedral electron-group arrangement.
Octahedral shape of SF_6 molecule.

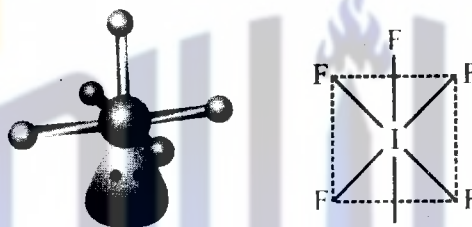
(2) Structure of Iodine Pentafluoride [IF_5]:

The outer electronic configuration of iodine ($_{53}\text{I}$) in the ground state is



In the excited state two electrons from $5p$ orbitals are excited to $5d$ orbitals, thereby giving five single electrons. $5s$, $5p_x$, $5p_y$, $5p_z$ and two d -orbitals hybridize to give six equivalent orbitals oriented octahedrally. Each of them is sp^3d^2 .

However one position is occupied by a lone pair. Therefore actual shape is **square pyramidal** Fig. (3.16)

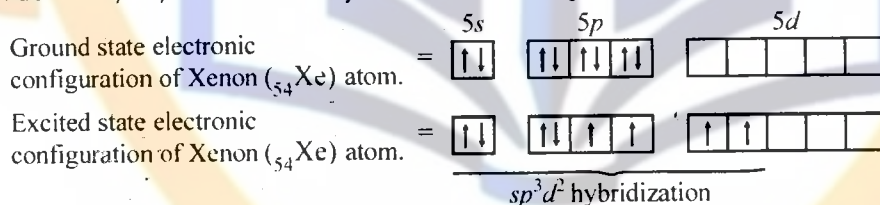


Square pyramidal

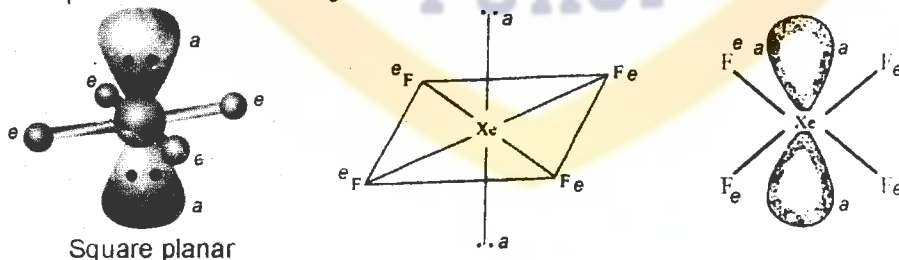
Fig. (3.16): Square pyramidal shape of IF_5 .

(3) Structure of Xenon Tetrafluoride XeF_4 :

The outer electronic configuration of Xe in the ground state is given. In the excited state, two of the $5p$ electrons go to the $5d$ -orbitals so that there are now four unpaired electrons. The central atom, Xe, is surrounded by six orbitals resulting from sp^3d^2 hybridization.



The hybrid orbitals are oriented octahedrally Fig. (3.17). Four of these orbitals contain shared pairs while the remaining two contain lone pair of electrons.



Square planar

Fig. (3.17): Different ways to sketch the square planar structures of XeF_4 molecule.

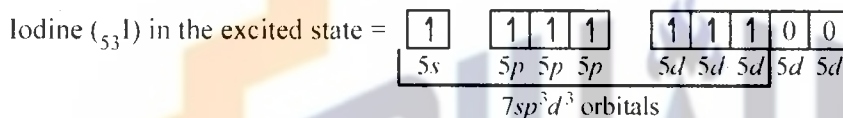
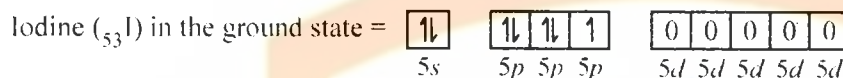
As the **vertex positions** are occupied by lone pairs, the actual shape is **square planar** as represented in Fig. (3.17)

3.4.6 sp^3d^3 Hybridisation

Here we mix one s , three p and three d atomic orbitals to give seven sp^3d^3 hybrid orbitals. They are directed towards the corners of a **pentagonal bipyramid**. These are not equivalent hybrid orbitals. Five of them point towards the vertices of a regular pentagon while others are perpendicular to the plane.

(1) Structure of Iodine Heptafluoride (IF_7):

The outer electronic configuration of iodine atom in the ground state and excited states are given.



To create half-filled orbitals for hybridization purposes, unpairing of 5s as well as of $5p_x$ and $5p_y$ electrons takes place. Thus seven unpaired electrons are available for bond formation to give rise to IF_7 molecule. The resulting molecule acquires **pentagonal bipyramidal** shape shown in Fig (3.18).

In this case all the seven hybrid orbitals are **not equivalent**. Five of them are directed towards the vertices of a **regular pentagon** giving an angle 72° with one another, the rest of two are oriented at right angle to the plane of the first set of five orbitals. Thus the bond angles in the molecule are of 72° and 90° .

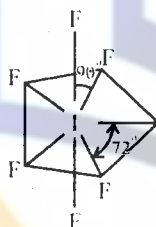


Fig. (3.18): Formation and structure of IF_7 molecule.

Similarly, the formation of XeF_6 is shown, which has irregular octahedral geometry.

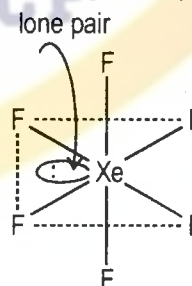
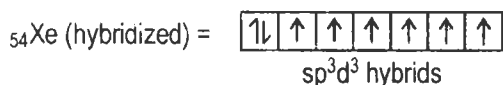
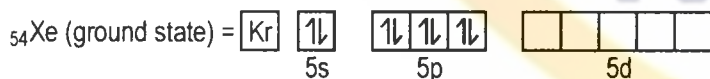


Fig. (3.19): Irregular octahedral geometry of XeF_6 .

3.4.7 AB_8 Type Molecules

d^4sp^3 Hybridization

As is clear that d^4sp^3 hybridization should involve the mixture of one s, three p and four d orbitals on the same atom. In this way, the atom forms eight identical d^4sp^3 hybrid orbitals. These eight hybrid orbitals result in three kinds of structures.

- Square antiprismatic,
- Dodecahedron

(a) Square Antiprismatic:

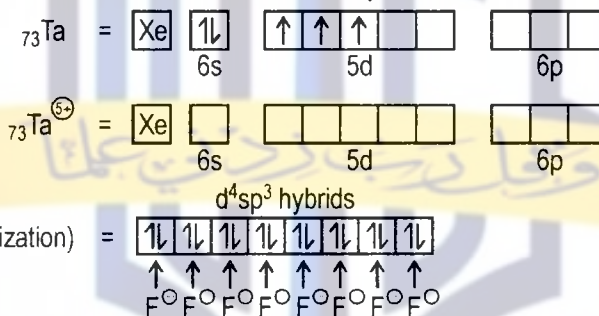
(i) The nature of the **central metal ion**, its **size**, **oxidation state**, **available orbitals**, **energy of the bonding orbitals** and

(ii) The nature of the ligand, its electronegativity, polarizability, crystal field splitting energy and its π bonding ability.

Let us discuss the geometry of $[\text{TaF}_8]^{3-}$ ion. It has square anti-prismatic structure.

Similarly $[\text{ReF}_8]^{2-}$ also has same structure.

Here, Ta is in +5 oxidation state. It is Ta^{5+} before hybridization.



Each F^- gives lone pair and 8 lone pairs are donated in eight vacant.

The geometry is shown in hybrid orbitals. Fig. (3.20)

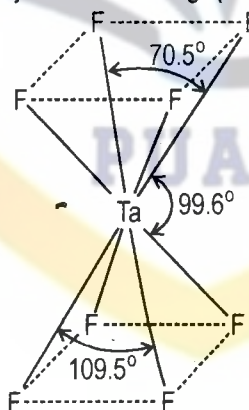


Fig. (3.20): Square anti-prismatic structure of $[\text{TaF}_6]^{3-}$ ion using d^4sp^3 hybridization.

(b) Dodecahedron Geometry:

These are two sides of coordination in dodecahedron geometry. Majority of the complexes studied so far of AB_8 types are dodecahedron. The following molecules adopt dodecahedron structure.

(i) These are complex molecules or ions which contain mixed ligands. The reason is that there are **two types of ligands, sides or coordination points**.

(ii) This geometry is favoured by **bidentate ligands**, e.g., $(\text{COO}^-)_2$, $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$. The reason is that there is less rigid situation in such cases.

$${}_{42}\text{Mo} = [\text{Kr}] \underset{4d}{\uparrow\uparrow\uparrow\uparrow} \underset{5s}{\uparrow\downarrow} \underset{5p}{}$$
$$_{42}\text{Mo}^{4+} = [\text{Kr}] \underset{4d}{\boxed{\uparrow\downarrow}} \underset{5s}{\boxed{}} \underset{5p}{\boxed{}}$$
[illegible]

d^4sp^3 hybrids

$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$
$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
CN	CN	CN	CN	CN	CN	CN	CN

(c) Hexagonal Bipyramide Geometry:

$$_{91}\text{Pa} = [\text{Rn}] \, 5f^2 6d^1 7s^2 7p^0$$
$${}_{91}\text{Pa}^{5+} = [\text{Rn}] \quad \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \quad \boxed{} \boxed{} \boxed{} \boxed{} \quad \boxed{} \boxed{} \boxed{} \quad \boxed{}$$

5f 6d 7p 8s

[illegible]

Table (3.2): Some Alternative and Less Common Hybridization Schemes.

Coordination Number	Arrangement of groups in three dimensional space	Scheme of Hybridization
2	Linear	pd, sd
3	Trigonal planar	p^2d
	Unsymmetrical	spd
	Trigonal pyramidal	pd^2
4	Tetrahedral	sd^3
	Irregular tetrahedral	spd^2, p^3d, pd^3
	Square planar	p^2d^2
5	Trigonal bipyramidal	spd^3
	Tetragonal pyramidal	$sp^2d^2, sd^4, pd^4, p^3d^2$
	Pentagonal planar	p^2d^3
6	Trigonal prism	spd^4, pd^5
	Trigonal antiprism	p^3d^3
7	Capped trigonal prism	d^4sp^2
8	Dodecahedral	d^4sp^3
	Square antiprismatic	d^5p^3
9	Tricapped trigonal prismatic	d^5sp^3

Limitations of Concept of Hybridization

Some limitations of the concept of hybridization are as follows:

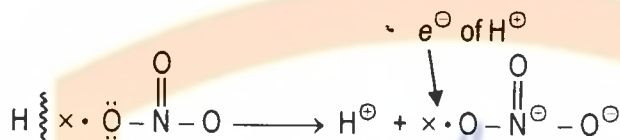
- (i) The promotion of electrons and hybridization of orbitals are necessary for formation of hybrid orbitals. However, a promotion of electron is not necessary for the formation of hybrid orbitals. This is particularly true in multiple bonding.
- (ii) It is believed that the driving force for hybridization is the energy released due to the bond formation of hybrid orbitals. We know that the carbon atom in CH_4 with ground state $2s^2 2p^2$ does the promotion of 2s electron to 2p orbital. It can do the sp^3 , sp^2 or sp hybridization of orbitals for bond formation. It depends upon the number of bonds to be formed in a particular situation. The energy consideration compel us to make sp^3 not sp^2 and sp.
- (iii) We assume that all five dsp^3 hybridized orbitals are equivalent. A trigonal bipyramidal hybridization gives two axial orbitals perpendicular to the three equatorial orbitals. Thus PCl_5 has two chlorine atoms different from the other three.

3.5 Shapes of Molecular Ions

(1) Structure of Nitrate Ion (NO_3^-):

In nitrate ion, nitrogen is the central atom and it has 5 electrons in outermost shell ($2s^2 2p^3$), while oxygen atom has six electrons ($2s^2 2p^4$). The nitrate ion carries a unit negative charge. It means that one of the oxygen atoms carries an extra electron and comes in the form of O^- to give NO_3^- .

Actually this extra electron is of H-atom. When HNO_3 (HONO_2) is dissociated it gives NO_3^- and H^+ ions.



The bond formation in nitrate ion can be shown in Fig. (3.24 a). There are two O-atoms and one O-ion. We write it three times around one N-atom in following diagram.

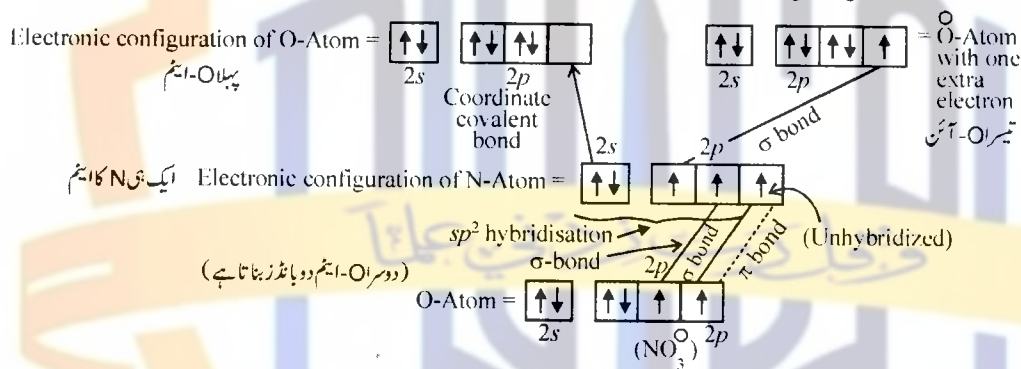


Fig. (3.24 a): Formation of nitrate ion from two neutral O-atoms and one negative oxygen.

The $2p_z$ orbital of nitrogen is left unhybridized and it is involved in π bond formation. In the rest of orbitals of N-atom one sp^2 has electron pair. Two have single electrons. Unhybridized $2p_z$ orbital of nitrogen has single electron.

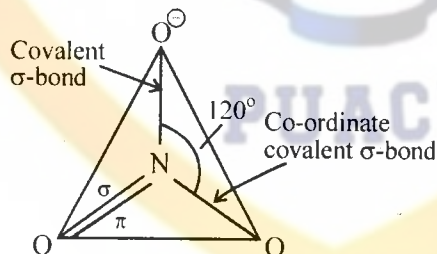


Fig. (3.24 b): Shape of NO_3^- ion which is *triangular planar*.

Total number of electron pairs used for hybridization after making NO_3^- are three and number of orbitals required are three i.e., one s and two p . So the hybridization is sp^2 . Thus the shape of molecule is **trigonal planar** with bond angle 120° as shown in Fig. (3.24 b).

(2) **Structure of Carbonate ion. (CO_3^{2-}):**

Electronic configuration of carbon is $1s^2 2s^2 2p^2$ in ground state while in excited state it is $1s^2, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$ as described below. We take one O-atom and two $O^\ominus$ -ions.

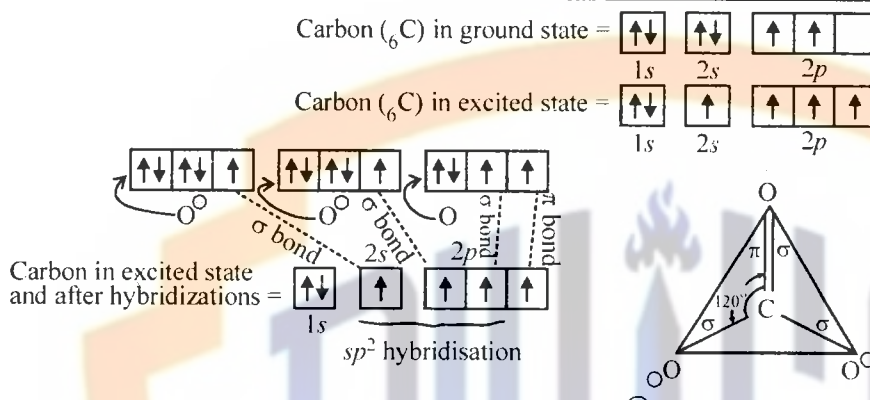


Fig. (3.25): Formation and structure of CO_3^{2-} ion.

In carbonate ion carbon atom shares four electrons forming four bonds. Out of these three are sigma (σ) and one pi (π) bond. Remember that π bonds are excluded from hybridization and therefore CO_3^{2-} ion has triangular planar shape Fig. (3.25).

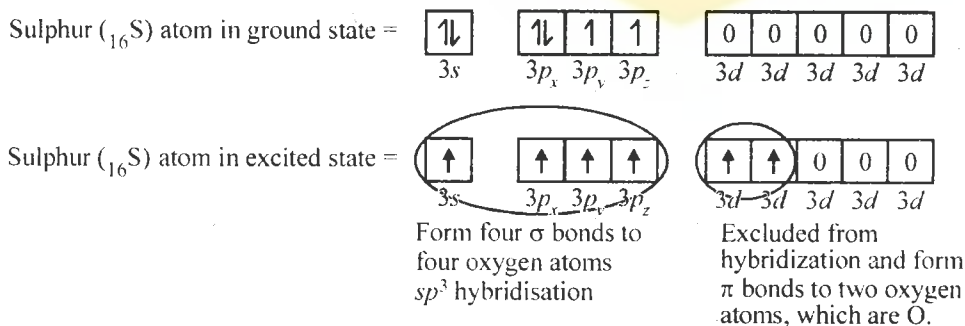
As carbonate ion has two negative charges. It is implied that **two oxygens have an extra electron while the third oxygen is normal**. The half-filled unhybridized orbital of carbon forms a π bond with that of oxygen atom having no extra electrons and thus bonded by a double-bond to carbon atom.

The bond angle is 120° . However carbon atom has no lone pair of electrons and hence the molecule has a regular geometry.

All bond lengths of C – O bonds are equal because double bond shows resonance.

(3) Structure of Sulphate Ion (SO_4^{2-}):

The outer electronic configuration of sulphur atom is $3s^2 3p^4$. The shape of molecule suggests that it is a case of sp^3 hybridization. To have the requisite number of half-filled orbitals one electron from s and one electron from p orbitals are excited to d -orbitals as shown in Fig. (3.26).



Two are O and
other two are O[∘].

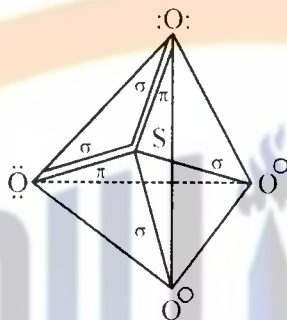


Fig. (3.26): Tetrahedral structure of SO_4^{2-} .

Out of four oxygen atoms two have extra electron and are O^- . That is why the net charge on SO_4^{2-} is -2 . In other words two oxygen atoms are neutral with two unpaired electrons each as



and two oxygen atom are O^{-1} . They have one unpaired electron in their p -orbitals which makes one π -bond. Sulphur combines with four oxygen atoms forming four bonds, thus using four pairs of σ -bond. Also sulphur atom has no non-bonded electron pair. The number of hybrid orbitals used are four viz., one s and three p orbitals as per sequence. The two half-filled d -orbitals form π bonds with two oxygen atoms, having no extra electron.

SO_4^{2-} has four resonance structures and all bond lengths are equal.

(4) Structure of Chlorate Ion (ClO_3^-):

Chlorine combines with three atoms of oxygen using three of its electrons. Thus it forms three single σ bonds with oxygen using three of its electrons. Here two electrons

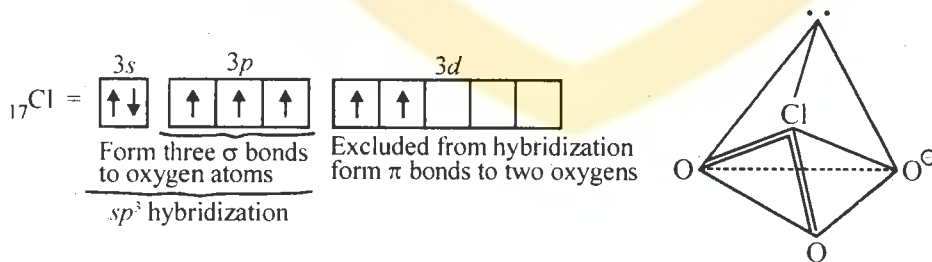


Fig. (3.27): Structure of Chlorate ion (ClO_3^-).

from $3p_x$ and $3p_y$ are excited to $3d$ orbitals which don't participate in hybridization but form π bonds with two oxygen atoms without extra electron. $3s$ and $3p$ orbitals of ${}_{17}\text{Cl}$ give sp^3 hybridization. It forms three single σ bonds with oxygen using three orbitals. Its actual shape is **pyramidal** with one corner position occupied by a lone pair of electrons. Its expected geometry is tetrahedral as sp^3 hybridization is involved. The two half-filled d -orbitals form π bonds with two oxygen atoms and thus bonded to chlorine by double bonds as shown in Fig. (3.27)

(5) Structure of Perchlorate ion (ClO_4^-):

The outer electronic configuration of Cl-atom is $3s^2 3p_x^1 3p_y^1 3p_z^1$. One electron each from $3s$, $3p_x$ and $3p_y$ are excited to $3d$ orbitals. Single electron in $3s$, $3p_x$, $3p_y$ and $3p_z$ hybridize in sp^3 manner to give tetrahedral shape. Electrons in $3d$ orbital form π bonds. ClO_4^- ion is tetrahedral. Chlorine has no non-bonded electron pair in the excited state. Three half-filled d -orbitals form π bonds with three oxygen atoms without extra electron and thus bonded by double bonds to chlorine as shown in Fig. (3.28).

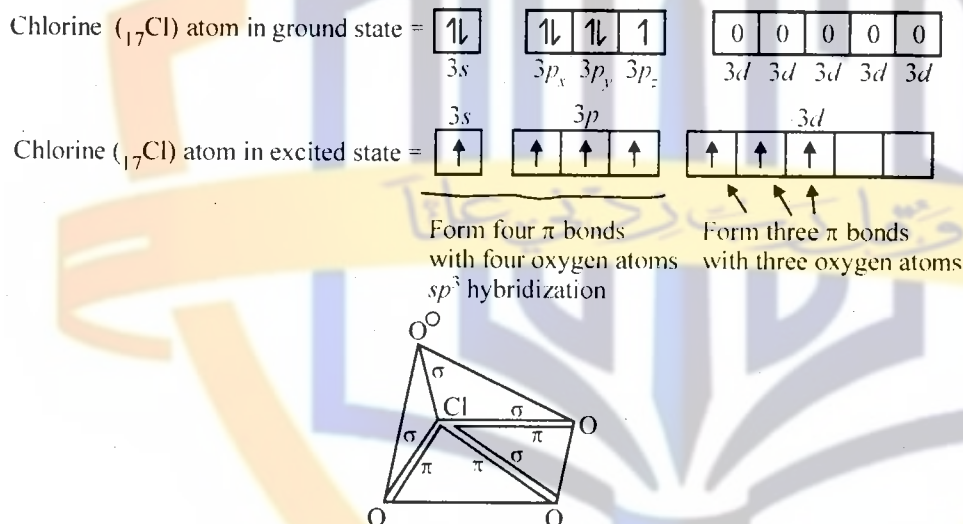


Fig. (3.28): Tetrahedral shape of perchlorate ion (ClO_4^-).

ClO_4^- shows four resonance structures and all bond lengths are equal.

3.6 Molecular Orbital Theory (MOT)

The main points of molecular orbital theory (MOT) put forward by F. Hund and R.S. Mulliken in 1930 are as follows.

- (i) Electrons belonging to two bonding atoms are controlled by the **entire molecule**. The electrons will occupy molecular orbitals which belong to the entire molecule.
- (ii) In an atom there are s , p , d atomic orbitals. These orbitals are determined by various values of quantum numbers. In a molecule we have a **set of molecular orbitals** σ , π , ...
- (iii) Like an atomic orbital, molecular orbital can contain a maximum of **two electrons**.
- (iv) The various electrons occupy different orbitals according to **Hund's rule** of maximum multiplicity as in the case of atomic orbitals.

3.6.1 Linear Combination of Atomic Orbitals (LCAO)

Here we need a very simple, reasonable and a satisfactory model of chemical bonding. It is obtained if it is assumed that atomic orbitals of constituents combine in a linear additive fashion to form molecular orbitals. Consider two atoms A and B. Each of them possesses a single valence electron. Let ψ_A and ψ_B represent the wave functions of atoms A and B. Now atomic orbitals A and B having valence electrons overlap. The valence electrons will be in reg-

ion where the orbitals overlap. Therefore, they can be associated with either nucleus. Consequently, the valence electrons have a much larger volume available for their movement as compared with that available in each isolated atom. The electrons can be thought of as moving in a molecular orbital whose total volume is approximately equal to the sum of the volume of atomic orbitals of A and B. In this way, the wave function of molecular orbitals can be expressed as a **linear combination of atomic orbitals (LCAO)**. ψ_{AB} is the wave function of molecular orbital and will be written as

$$\psi_{AB} = C_A \psi_A + C_B \psi_B$$

Here C_A and C_B are the mixing coefficients. Value of C_A and C_B is so chosen as to get a molecular orbital of lowest energy.

When we consider the bonding in two similar atoms, then the magnitude of the coefficients C_A and C_B must be same.

In that case

$$C_A^2 = C_B^2 \text{ or } C_A = \pm C_B$$

In this way there will be two molecular orbitals as a result of two types of combinations as given below.

$$\psi^b = \psi_A + \psi_B$$

$$\psi^* = \psi_A - \psi_B$$

If values of C_A and C_B are taken as unity

One molecular orbital (ψ^b) is obtained by addition of wave function of two atomic orbitals. ψ^b stands for bonding. The other molecular orbital (ψ^*) is obtained by the subtraction of wave function of the atomic orbitals.

As discussed earlier that the **number of molecular orbitals is always equal to the number of atomic orbitals involved in bonding**. Effective combination of atomic orbitals to give rise to stable molecular orbitals requires:

- (i) **Symmetry** and **relative energies** of atomic orbitals, ψ_A and ψ_B must be close to each other.
- (ii) **Extent of overlap** between the charge clouds of ψ_A and ψ_B should be large.
- (iii) Atomic orbitals of **same sign should overlap**.

3.6.2 Bonding and Anti-bonding Molecular Orbitals

The molecular orbital that is obtained by the addition overlap of two atomic orbitals is termed as bonding molecular and is represented as ψ^b .

$$\psi^b = \psi_A + \psi_B$$

Some features of bonding molecular orbitals (BMO) are:

- (i) BMO possesses **lower energy** than the atomic orbitals from which it is formed.
- (ii) BMO has **high electron density** in the region between the two nuclei and this is the reason for stability of bond.
- (iii) BMO is formed when lobes of the combining atomic orbitals have **same sign**. The magnitude of ψ^b is numerically bigger than ψ_A or ψ_B .

(iv) Every electron in a BMO contributes to **attraction** between two atoms.

Anti-bonding Molecular Orbital (ABMO):

The molecular orbital obtained by the subtraction of overlap of two atomic orbitals is termed the anti-bonding molecular orbital and is represented as ψ^ .*

$$\psi^* = \psi_A - \psi_B$$

Some features of ABMO are:

- (i) ABMO possesses **higher energy** than the atomic orbitals from which it is formed.
- (ii) The **probability** of finding the electron in between the nuclei is **negligible** in ABMO.
- (iii) ABMO is formed when the lobes of combining atomic orbitals have **opposite signs**. ψ^* is numerically smaller than ψ_A or ψ_B .
- (iv) Every electron in an ABMO contributes to **repulsion** between two atoms.

3.6.3 Combination of s Orbitals to Give Molecular Orbitals (MO)

Consider two hydrogen atoms with their s-orbitals. They have one electron each. Using LCAO approach the two atomic orbitals will overlap to give two molecular orbitals. The first combination is obtained by the addition of two atomic orbitals. The pictorial representation is shown in Fig. (3.24)

$\psi^b \text{ MO} = N (\psi_{1s_a} + \psi_{1s_b})$ where N is a normalizing constant. The value of N may be taken as 1 for simplicity.

The electron in this MO will spend most of its time between the two nuclei H_a and H_b . The electron is now stabilized by two nuclear charges instead of one in an atomic orbital. The resulting MO is of lower energy than the 1s orbital. That is why it is called a BMO.

The second possibility of linear combination is obtained by subtracting the two atomic orbitals and is shown in Fig. (3.29)

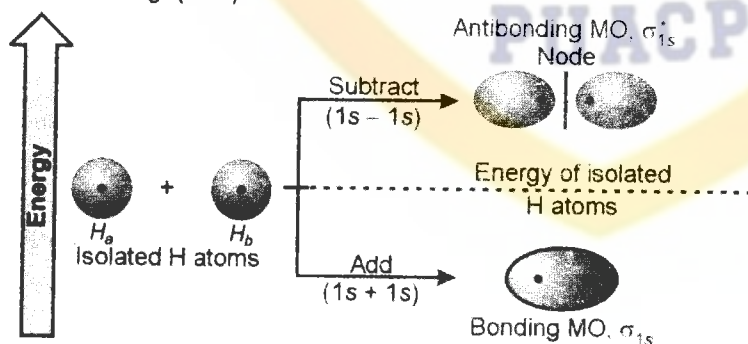


Fig. (3.29): Bonding and antibonding M.O due to s-s overlap.

The resulting molecular orbital has zero electron density between the nuclei. The region of zero electron density is called **nodal plane**. The net effect is **to push the nuclei apart** rather than to hold them together. The resulting molecular orbital is of **higher energy** than the isolated 1s atomic orbital. The molecular orbitals discussed above have **cylindrical symmetry** and are called σ (sigma) molecular orbitals just like the combination of two 1s atomic orbitals.

The combination of 2s atomic orbitals also gives rise to σ -molecular orbitals. The only difference is that their sizes are different.

3.6.4 Formation of σ Molecular Orbitals From p -orbitals Overlapping along the Internuclear Axis

Sigma type molecular orbitals are also formed when p_z atomic orbitals of two atoms combine with each other. We are now assuming that the internuclear axis is the z -axis. This is illustrated in a diagram shown in Fig. (3.30)

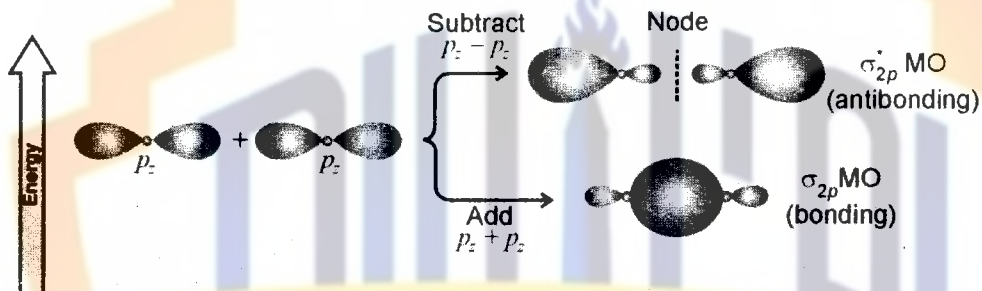


Fig. (3.30): Formation σ M.O due to head on overlapping of two p -orbitals.

The molecular orbital formed by the addition, overlap of $2p_z$ atomic orbital is designated as $\sigma_{2p_z}^b$ or simply as σ_z^b . The molecular orbital formed by the subtraction of $2p_z$ atomic orbitals is designated as $\sigma_{2p_z}^*$ or simply σ_z^* . These orbitals also have cylindrical symmetry around the molecular axis and hence they are also classified as σ -molecular orbitals Fig. (3.25)

3.6.5 Formation of π Orbitals

Those molecular orbitals which are obtained by the lateral or sidewise overlap of p_x or p_y orbitals are designated as π (pi) molecular orbitals. This is illustrated by showing overlap of $2p_x$ atomic orbitals.

Overlaps containing the same algebraic signs yield BMO and those of opposite sign give ABMO Fig. (3.31).

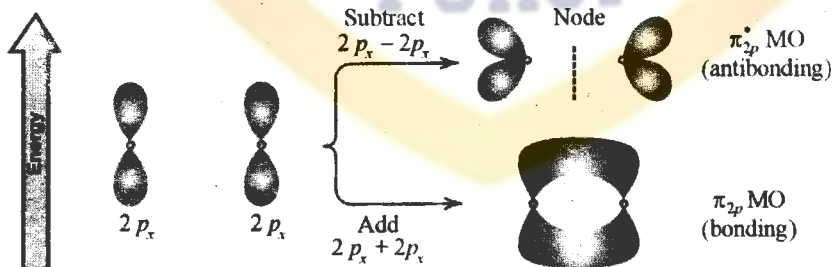


Fig. (3.31): Formation of π M.O due to parallel overlapping of two p_x -orbitals.

The bonding molecular has (+) lobe above the z -axis and (–) lobe below the z -axis. The plus and minus signs refer to the sign of wave function in the various lobes and not to nuclear or electronic charges. The (+ve) sign or upward direction represents one phase and (–ve) sign or downward direction represents second phase of wave function.

The two $2p_y$ atomic orbitals will overlap in a similar fashion to give π_y^b and π_y^* . Both π_x and π_y molecular orbitals are equivalent and have the same energy. These orbitals are said to be degenerate.

Table (3.3): Comparison of BMO and ABMO.

Bonding Molecular Orbitals (BMO)	Anti-bonding Molecular Orbitals (ABMO)
(i) Bonding molecular orbitals (BMO) are obtained when two atomic orbitals combine together in such a way that their lobes with ++ sign overlap. It is a ++ overlap .	(i) Anti-bonding molecular orbitals (ABMO) are obtained when two atomic orbitals combine together in such a way that their lobes with + - sign overlap. It is a + - overlap .
(ii) The probability of finding the electron in the region between the nuclei of the combining atoms is very high and hence a stable molecule is formed.	(ii) The probability of finding the electron in the region between the nuclei of the combining atoms is negligible .
(iii) The energy of bonding molecular orbital is less than that of both the combining atomic orbitals and hence the formation of bonding molecular orbital leads to the formation of a stable bond.	(iii) The energy of anti-bonding molecular orbital is more than that of both the combining atomic orbitals and hence the formation of an anti-bonding molecular orbital produces repulsion between the combining atoms.
(iv) Each electron in a bonding molecular orbital contributes to attraction between the two combining atoms.	(iv) Each electron in an anti-bonding molecular orbital contributes to repulsion between the combining atoms.
(v) The difference in energy between the molecular orbital and the combining atomic orbitals is called stabilization energy .	(v) The difference in energy between the molecular orbital and the combining atomic orbitals is called destabilization energy .

Table (3.4): Comparison Between Sigma and pi Molecular Orbitals.

Sigma (σ) molecular orbitals	Pi (π) molecular orbitals
(i) σ MO is obtained when ++ or +- portions of two atomic orbitals overlap with each other along the internuclear axis .	(i) π MO is obtained when ++ or +- portions of the two atomic orbitals overlap with each other along a line perpendicular to the internuclear axis .
(ii) It has cylindrical symmetry around the internuclear axis. It is for this reason that these molecular orbitals are called sigma molecular orbitals.	(ii) This is concentrated perpendicular to the molecular axis, these molecular orbitals do not have cylindrical symmetry around the molecular axis.
(iii) It can be rotated freely about the bond axis.	(iii) It cannot be rotated freely about the bond axis.
(iv) It is a stronger molecular orbital, since it results by the greater overlap of orbitals along the nuclear axis.	(iv) It is a weaker molecular orbital since it results by the lesser overlap of orbitals along a line perpendicular to the molecular axis.

(v) The formation of a σ -molecular orbital decides the direction and length of the bond.	(v) The formation of a π -molecular orbital does not decide the direction of the bond but shortens the bonds distance. The formation of two π -molecular orbitals shortens the bond distance to a greater extent than the formation of one π -molecular orbital.
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3.6.6 Rules for Filling Electrons in Molecular Orbitals

Following rules are followed for filling of electrons in different molecular orbitals.

- (i) Electrons enter molecular orbitals in the order of their increasing energies. The electron will first enter the lowest energy molecular orbital according to **Aufbau Principle**.
- (ii) Each molecular orbital can accommodate a maximum of **two electrons** with their spins in **opposite direction** in accordance with **Pauli's exclusion principle**.
- (iii) According to **Hund's rule** whenever two equal energy molecular orbitals are available, electrons **prefer to remain unpaired**.
- (iv) Electrons are placed one at a time into molecular orbital to obtain the electronic configuration of the molecule.

3.6.7 Energy Level Diagram of Molecular Orbitals

Energy level diagram of MO's is shown in Fig. (3.32)

In the diagram the combining atomic orbitals are shown on two extremes. The resulting molecular orbitals are represented in the middle column. Atomic orbitals of same energy are shown at the same level. **The difference between the energies of atomic orbitals and the bonding MO's depend upon the extent of overlap of atomic orbitals.**

A large overlap will result in greater lowering of BMO and a corresponding large raise in the ABMO. Larger the overlap, greater is the overlapping and hence stronger will be resulting bond.

According to following diagram the sequence of energy levels in which they are filled up is:

$$\sigma_{1s}^b < \sigma_{1s}^* < \sigma_{2s}^b < \sigma_{2s}^* < \sigma_x^b < \pi_y^b = \pi_z^b < \pi_y^* = \pi_z^* < \sigma_x^*$$

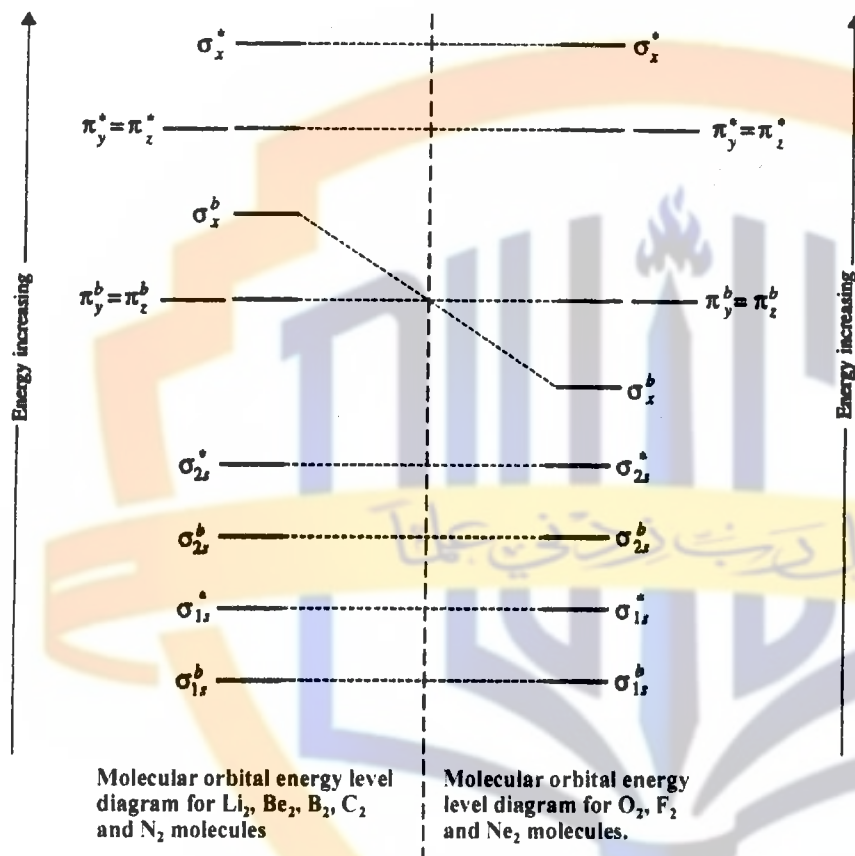


Fig. (3.32): Molecular orbital energy level diagram for Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2 , F_2 and Ne_2 molecules.

The sign of star * represents the antibonding situation.

MO energy levels for O_2 , F_2 , and Ne_2 are shown Fig. (3.32). The six atomic $2p$ orbitals of the two atoms form six MOs. Those forming π orbitals yield two bonding MOs (π_{2p}) of equal energy and two antibonding MOs (π_{2p}^*) of equal energy.

MO energy levels for B_2 , C_2 , and N_2 are shown Fig. (3.32). Because of significant ($\text{جداً$) $2s$ - $2p$ orbital mixing, the energies of σ MOs formed from $2p$ orbitals increase and those formed from $2s$ orbitals decrease. The major effect of such orbital mixing on the MO sequence is to make the σ_{2p_x} higher than the π_{2p_y} and π_{2p_z} .

The sequence in case of B_2 , C_2 and N_2 is as follows.

$$\pi_y^b = \pi_z^b < \sigma_x^b < \pi_y^a = \pi_z^a < \sigma_x^a$$

The sequence can be seen in Fig. (3.32).

The sequence (ترتيب) of energies mentioned above does not apply to heavier elements like O_2 and F_2 and Ne_2 . Here $2s - 2p$ mixing takes place which changes the sequence of energy.

So, $\sigma_{1s}^b < \sigma_{1s}^a < \sigma_{2s}^b < \sigma_{2s}^a < \sigma_x^b < \pi_y^b = \pi_z^b < \pi_y^a = \pi_z^a < \sigma_x^a$ (Is true for O_2 , F_2)

Lower molecular orbitals have the same energy sequence.

3.6.8 Bond Order from Number of Electrons in Bonding and Antibonding Molecular Orbitals

When all the electrons occupying bonding molecular orbitals (N^b) and antibonding molecular orbitals (N^*) are counted separately the following cases arise.

- If the number of electrons in BMO's are greater than the number of electrons in ABMO's then the molecule would be stable.
- If the numbers of electrons in BMO's are equal to the number of electrons in ABMO's then the molecule is unstable and it does not exist.

Bond Order:

It is defined as the number of covalent bonds present in a molecule. It is calculated as

$$\text{Bond order} = \frac{\text{Number of electrons in BMO} - \text{Number of electrons in ABMO}}{2}$$

Information Conveyed by Bond Order:

- The bond order gives **nature of a bond** whether it is a single, double or a triple.
- When the bond order comes out to be **zero**, then there is no possibility of bond formation.
- As the bond order increases, bond length decreases.
- Bond dissociation energy** is directly proportional to the bond order for diatomic molecules.

3.6.9 Molecular Energy Level Diagram for Hydrogen Molecule (H_2)

H_2 is formed by the overlap of $1s$ atomic orbitals. It shall have two molecular orbitals.

One is bonding (σ_{1s}^b) and the other is antibonding (σ_{1s}^*). The H_2 molecule has two electrons and these can be accommodated in the σ_{1s}^b molecular orbital. In accordance with **Pauli exclusion principle**, these electrons must have opposite spins. The energy level diagram of H_2 is shown in Fig. (3.33)

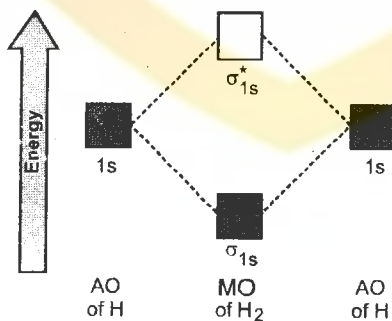


Fig. (3.33): The MO diagram for H_2 . The diagram indicates the relative energies and electron occupancy of the MOs and the AOs from which they are formed. (AO stands for atomic orbital)

The ground state of H_2 is represented as $[(\sigma_{1s}^b)^2]$ and the bond order is $1/2 (2 - 0) = 1$, i.e., one σ -bond. Bond dissociation energy in H_2 is $431.4 \text{ kJ mol}^{-1}$ and bond length is 74 pm .

(1) **Molecular Orbital Energy Level Diagram for $\text{H}_2^{\oplus}$:**

The $\text{H}_2^{\oplus}$ molecule ion will have the electronic configuration $[(\sigma_{1s}^b)^1]$

The bond order in $\text{H}_2^{\oplus} = \frac{1}{2} [1 - 0] = 1/2$ σ bond

Since the bond order in $\text{H}_2^{\oplus}$ is only one-half of a normal covalent bond, it will have low bond dissociation energy and large bond length.

Bond dissociation energy of $\text{H}_2^{\oplus} = 255 \text{ kJ mol}^{-1}$. Bond length = 106 pm.

(2) **Molecular Orbital Energy Level Diagram for He_2 Molecule:**

For He_2 molecule to be formed by the combination of two helium atoms, a total of four electrons are involved. From a combination of two $1s$ atomic orbitals we have two molecular orbitals σ_{1s}^b and σ_{1s}^* . Two electrons would go in the bonding σ_{1s}^b MO and the other two would have to go into the antibonding σ_{1s}^* MO Fig. (3.34).

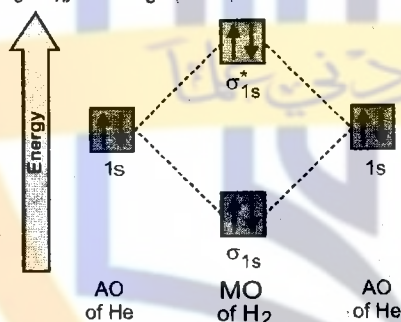


Fig. (3.34): MO diagrams for He_2 . In He_2 , the four valence electrons fill both the σ_{1s} and the σ_{1s}^* MOs. (AO stands for atomic orbital)

The electronic configuration of He_2 may be written as $[(\sigma_{1s}^b)^2 (\sigma_{1s}^*)^2]$.

The bond order is $1/2 [2 - 2] = 0$ in fact the bonding and anti-bonding orbitals cancel each other and there is no existence of He_2 molecule.

(3) **Molecular Energy Level Diagram in $\text{He}_2^{\oplus}$:**

Here three electrons are to be filled in the molecular orbitals Fig. (3.35). The configuration is $(\sigma_{1s}^b)^2 (\sigma_{1s}^*)^1$. This is an hypothetical ion.

The bond order is: $\frac{1}{2} [2 - 1] = \frac{1}{2}$.

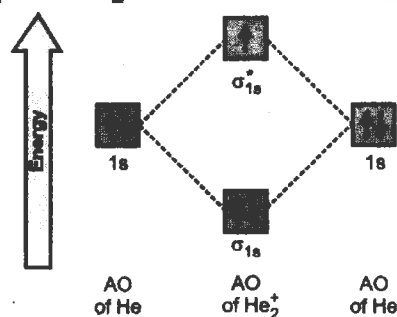


Fig. (3.35): MO diagram for $\text{He}_2^{\oplus}$.

(4) Energy Level Diagrams of Li_2 , Be_2 and B_2 :MO Diagram for Li_2 .

MOs obtained from 1s orbitals combination are lower in energy than those obtained from 2s orbitals. The six electrons from two Li atoms fill the lowest three MOs, and the σ_{2s}^* remains empty. With a bond order of 1, Li_2 does form. Fig. (3.36 a)

MO Diagram for Be_2 .

The eight electrons from two Be atoms fill all the available MOs to give no net stabilization. Ground-state Be_2 has a zero bond order and has never been observed.

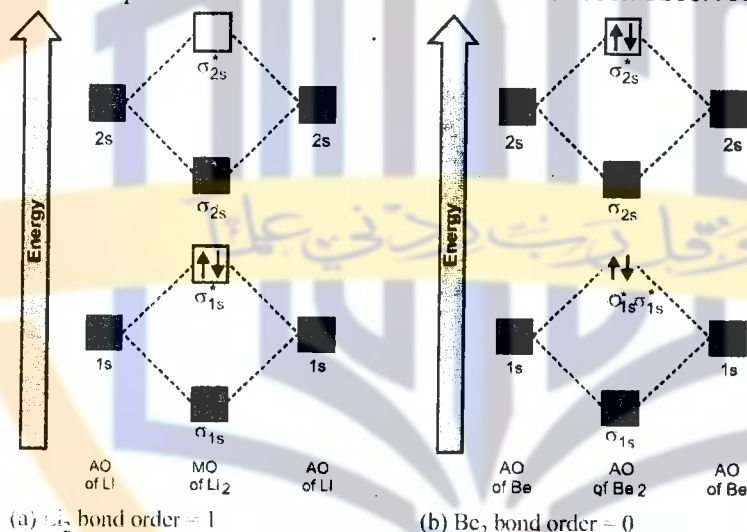


Fig. (3.36): (a) MO diagram for Li_2 , (b) MO diagram for Be_2

MO Diagram for B_2 .

When p - s mixing is allowed, the energies of the σ_{2p} and π_{2p} orbitals are reversed. The two electrons from the p orbitals of B, now occupy separate, degenerate π_{2p} molecular orbitals and thus have parallel spins. Therefore, this diagram explains the observed paramagnetism of B_2 . The bond order is 1 Fig. (3.37)

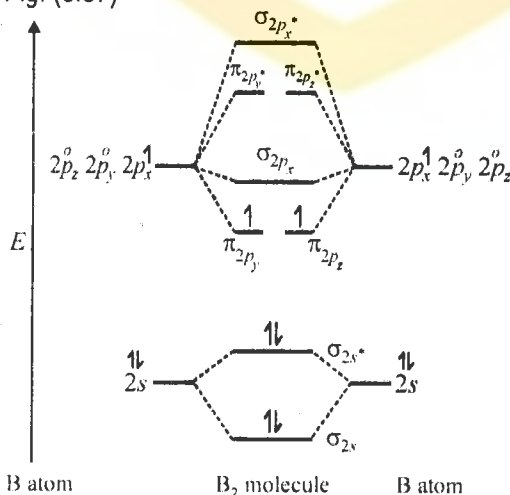


Fig. (3.37): Molecular orbital energy level diagram for the B_2 molecule.

(5) **Energy Level Diagram for the Formation of N₂ Molecule:**

The electronic configuration of N-atom is $1s^2, 2s^2, 2p^3$. Its molecular orbital configuration may be written as $[KK (\sigma_{2s}^b)^2 (\sigma_{1s}^*)^2 (\pi_y^b)^2 = (\pi_z^b)^2 (\sigma_x^b)^2]$. KK signifies that the K shells of the combining atoms are full and it includes σ_{1s} and σ_{1s}^* . N₂ has eight electrons in bonding and two electrons in antibonding orbitals. Since there are six electrons in excess in bonding orbitals so the bond order in N₂ is three. The molecule has one σ and two π bonds. This is in conformity with its very **high bond energy of 940 kJ mol⁻¹** and very short bond length of **110 nm**.

The energy level diagram Fig. (3.38) is shown below.

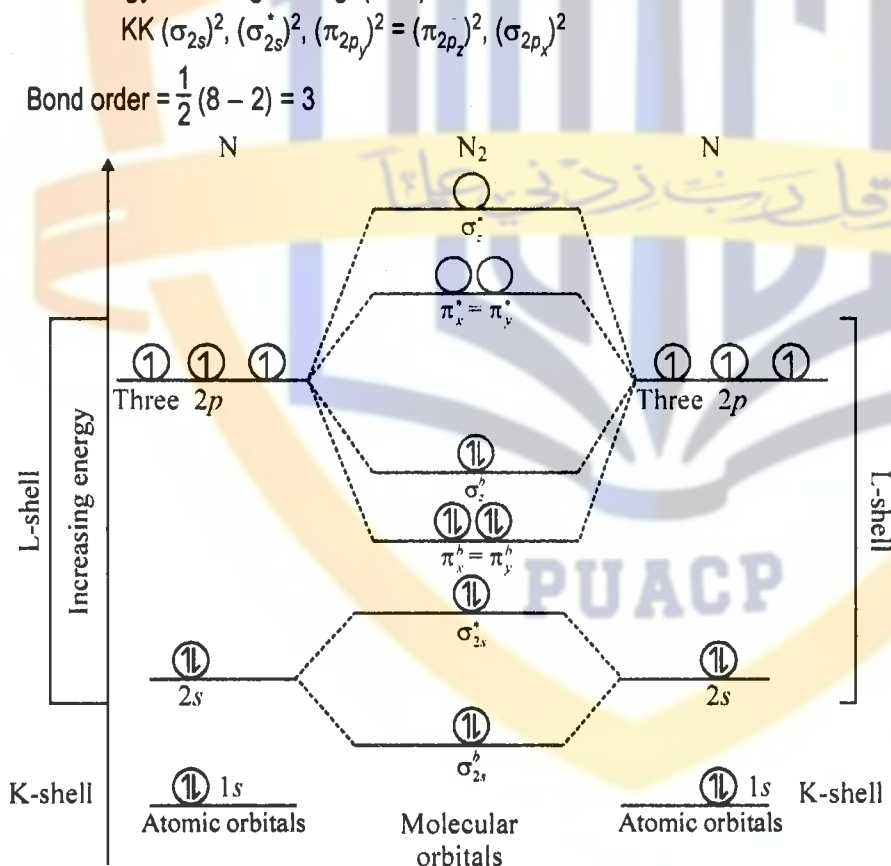


Fig. (3.38): Molecular orbital energy level diagram of hypothetical N₂ molecule.

(6) **Molecular Orbital Energy Level Diagram for (O₂) Molecule:**

The electronic configuration of oxygen is $1s^2, 2s^2, 2p_x^1 2p_y^1 2p_z^1$. The molecule has the following configuration:

$$[KK (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_{2p_x})^2 = (\pi_y^b)^2 = (\pi_z^b)^2, (\pi_y^*)^1 = (\pi_z^*)^1 \sigma_{2p_x}^*]$$

[One K stands for σ_{1s} and other for σ_{1s}^*]

$$\text{Bond order} = \frac{1}{2} (8 - 4) = \frac{1}{2} \times 4 = 2$$

Hund's rule says that the two electrons in the π^* orbitals should have parallel spins. The molecular orbital method explains the presence of two unpaired electrons in the molecule. **These two unpaired electrons account for paramagnetism of O_2 . The valence bond theory failed to predict the presence of unpaired electrons in O_2 molecule** Fig. (3.39)

This is in accordance with the experimental results that O_2 consists of a double bond (one σ and one π bond).

Bond dissociation energy of O_2 molecules is $495.2 \text{ kJ mol}^{-1}$ and bond length is 121 pm.

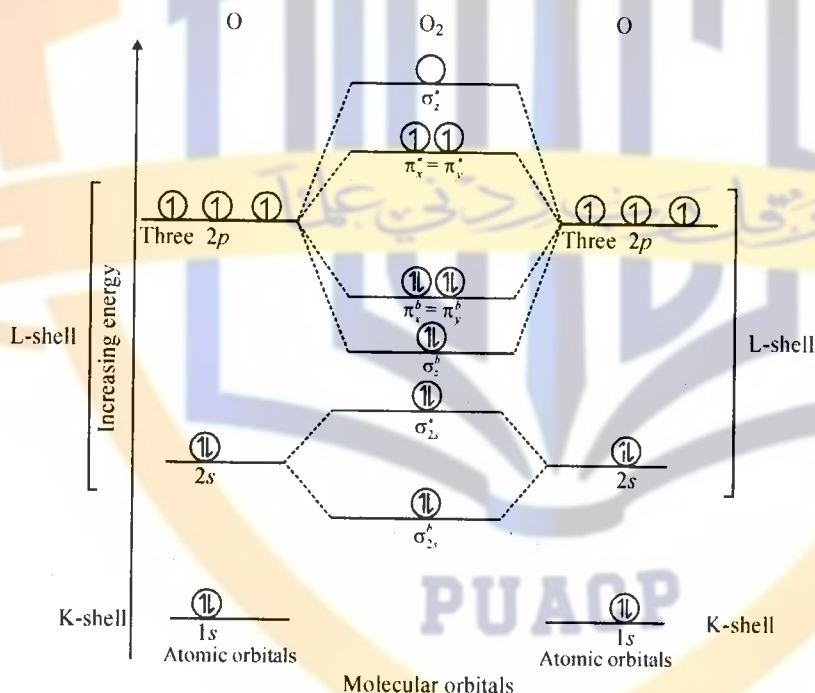


Fig. (3.39): Molecular orbital energy level diagram of O_2 .

(7) MO Energy Level Diagram of Fluorine (F_2) Molecule:

If we add two more electrons to the O_2 configuration, we get configuration for fluorine (F_2) molecule. This is shown in Fig. (3.40) The molecular orbital configuration of F_2 is:

$$(\sigma_{1s}^b)^2, (\sigma_{1s}^*)^2, (\sigma_{2s}^b)^2, (\sigma_{2s}^*)^2, (\sigma_x^b)^2, (\pi_z^b)^2 = (\pi_y^b)^2, (\pi_x^b)^2 = (\pi_y^*)^2, (\sigma_x^*)^0$$

and its bond order is $1/2 [10 - 8] = 1$. It is diamagnetic since it has no unpaired electrons.

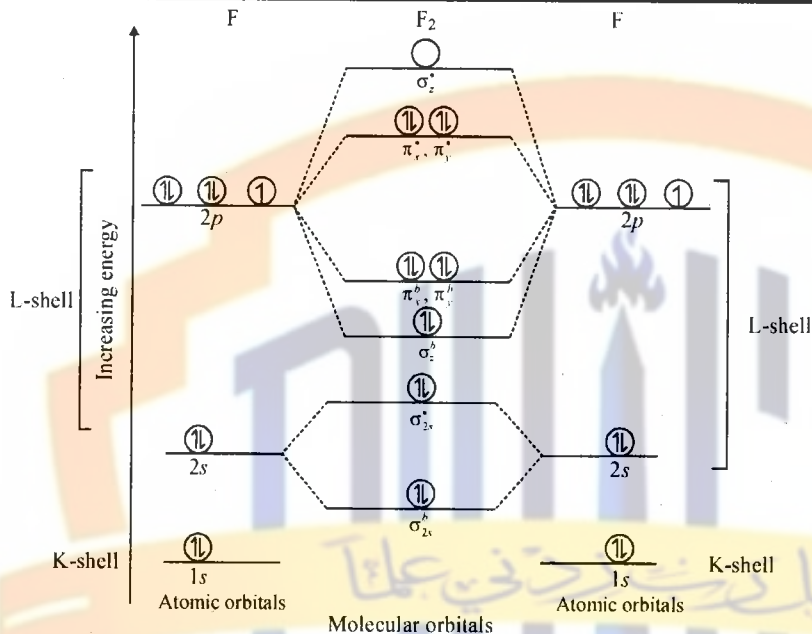
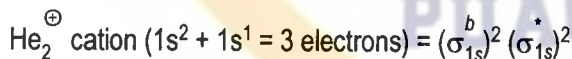
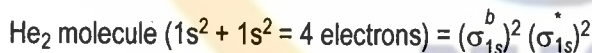
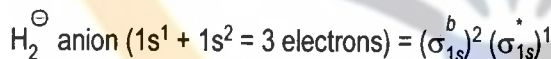
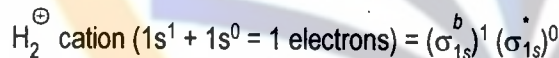
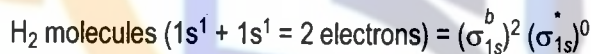


Fig. (3.40): Molecular orbital energy level diagram of F_2 .

The molecular orbital configurations of the given species can be written as:

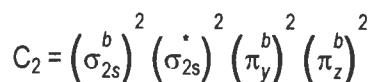


Molecular Orbital Configuration of C_2 Molecule:

Configuration of the valence-shell of C-atom is $2s^2, 2p_x^1, 2p_y^1$. The overlap of four atomic orbitals of one C-atom with the identical four atomic orbitals of the other C-atom gives eight molecular orbitals which are $\sigma_{2s}^b, \sigma_{2s}^*, \pi_y^b, \pi_z^b, \sigma_x^b, \pi_y^*, \pi_z^*, \sigma_x^*$. The relative order of the energy of these molecular orbitals is

$$\sigma_{2s}^b < \sigma_{2s}^* < \pi_y^b = \pi_z^b < \sigma_x^b < \pi_y^* = \pi_z^* < \sigma_x^*$$

Total number of electrons which occupy the molecular orbitals is eight. Thus the molecular orbital configuration of C_2 molecule is as:



This shows that bond order of C_2 molecule is $(6 - 2)/2 = 2$. It means that the two C-atoms in C_2 molecule are linked together by a double bond ($:C = C:$). Since C_2 molecule does not have any unpaired electrons, it is **diamagnetic**. It has been observed experimentally.

Bond dissociation energy of C_2 molecule is $627.9 \text{ kJ mol}^{-1}$ and bond length is 131 pm . These are experimental values.

The molecular orbital configurations of N_2 and $N_2^{\oplus}$ are as follows:

$$N_2 \text{ molecule (10 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\pi_y^b)^2 (\pi_z^b)^2 (\sigma_x^b)^2$$

$$N_2^{\oplus} \text{ ion (9 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\pi_y^b)^2 (\pi_z^b)^2 (\sigma_x^b)^1$$

Just like N_2 M.O picture of $N_2^{\oplus}$ ion can also be sketched.

The bond orders of N_2 molecule $= (8 - 2)/2 = 3$ and B.O. of $N_2^{\oplus}$ ion $= (7 - 2)/2 = 2.5$. Thus the bond order of N_2 is greater than that of $N_2^{\oplus}$. The N – N bond length in $N_2^{\oplus}$ ion $= 112 \text{ pm}$ and dissociation energy of N_2 is 940 kJ mol^{-1} . For $N_2^{\oplus}$ $= 828 \text{ kJ mol}^{-1}$. Thus N_2 molecule is more stable than $N_2^{\oplus}$ ion.

N_2 molecule has all the electrons in the paired state and hence is **diamagnetic**. On the other hand $N_2^{\oplus}$ ion has one unpaired electron and hence is **paramagnetic** to the extent of one unpaired electron. The bond order of 2.5 of $N_2^{\oplus}$ ion guides us that it has two normal covalent bonds and one 3-electron bond. Thus $N_2^{\oplus}$ ion can be represented by the structure $[N \equiv N:]^{\oplus}$.

Comparison of Molecular Orbital Configurations of O_2 , $O_2^{\oplus}$ and $O_2^{2\oplus}$ species:

($O_2 = 6 + 6 = 12$ electrons, $O_2^{\oplus} = 12 - 1 = 11$ electrons, $O_2^{\ominus} = 12 + 1 = 13$ electrons and $O_2^{2\oplus} = 12 + 2 = 14$ electrons). The molecular orbital configurations of these species are given below:

$$O_2 \text{ molecule (12 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_x^b)^2 (\pi_y^b)^2 = (\pi_z^b)^2 (\pi_y^*)^1 = (\pi_z^*)^1$$

$$\text{Oxygenyl ion, } O_2^{\oplus} \text{ (11 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_x^b)^2 (\pi_y^b)^2 = (\pi_z^b)^2 (\pi_y^*)^1 = (\pi_z^*)^0$$

$$\text{Superoxide ion, } O_2^{\ominus} \text{ (13 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_x^b)^2 (\pi_y^b)^2 = (\pi_z^b)^2 (\pi_y^*)^2 = (\pi_z^*)^1$$

$$\text{Peroxide ion, } O_2^{2\oplus} \text{ (14 electrons)} = (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_x^b)^2 (\pi_y^b)^2 (\pi_z^b)^2 (\pi_y^*)^2 = (\pi_z^*)^2$$

Molecular orbital energy level diagram of O_2 molecule is shown in Fig. In a similar manner these diagrams for other species can also be drawn.

Peroxide ion, $O_2^{2\oplus}$ has bond order as $(8 - 6)/2 = 1$ which shows that in this ion the two O-atoms are linked by one 2-electron bond. Thus the structure of this ion can be shown as $[:\ddot{O} - \ddot{O}:]^{2\oplus}$. Since this structure has no unpaired electron, this ion is **diamagnetic**. Bond ord-

er equal to one explains the low value of O – O bond dissociation energy ($210.0 \text{ kJ mol}^{-1}$) and O – O bond length = 149 pm. Following Table (3.5) summarizes the various parameters.

Table (3.5): Properties of O_2 , $\text{O}_2^{\ominus}$, $\text{O}_2^{\ominus}$ and $\text{O}_2^{\ominus}$
(B.O. = Bond order, n = No. of unpaired electrons).

S. No.	Species	Bond order	Structure	Unpaired electrons and magnetic property	O – O bond dissociation energy (kJ mol^{-1})	O – O bond length (pm)
1	(i) O_2 molecule (12)	$8 - 4/2 = 2 = 1 + 2 \times 0.5 = 2$	$:\text{O} \equiv \text{O}:$	2 (paramagnetic)	494	121
2	(ii) Oxygenyl ion, $\text{O}_2^{\ominus}$ (11)	$8 - 3/2 = 2.5$ ($2 + 1 \times 0.5$) = 2.5	$[\text{O} \equiv \text{O}]^{\ominus}$	1 (paramagnetic)	642	112
3	(iii) Superoxide ion, $\text{O}_2^{\ominus}$ (13)	$8 - 5/2 = 1.5$ ($1 + 1 \times 0.5$) = 1.5	$[\text{O} \equiv \text{O}]^{\ominus}$	1 (paramagnetic)	394	125
4	(iv) Peroxide ion, $\text{O}_2^{\ominus}$ (14)	$8 - 6/2 = 1$	$[\ddot{\text{O}} - \ddot{\text{O}}]^{\ominus}$	0 (diamagnetic)	210	149

Conclusions:

- The number of unpaired electrons (n) is equal to the number of 3-electron bonds.
- Since O_2 molecule has the maximum number of unpaired electrons, it has the maximum paramagnetic character.
- Bond order, bond dissociation energy and stability of the four species are in the order $\text{O}_2^{\ominus} > \text{O}_2 > \text{O}_2^{\ominus} > \text{O}_2^{\ominus}$ while the O – O bond length is in the order $\text{O}_2^{\ominus} < \text{O}_2 < \text{O}_2^{\ominus} < \text{O}_2^{\ominus}$.

Molecular orbital configuration of Ne_2 molecule ($8 + 8 = 16$ electrons). The valence-shell configuration of Ne-atom is $2s^2 2p_x^2 2p_y^2 2p_z^2$ (eight electrons). Ne_2 molecule has $8 + 8 = 16$ electrons which are accommodated in eight molecular orbitals. Thus the molecular orbital configuration of Ne_2 molecule is,

$$(\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\sigma_x^b)^2 (\pi_y^b)^2 (\pi_z^b)^2 (\pi_y^*)^2 (\pi_z^*)^2 (\sigma_x^*)^2$$

The bond order is equal to $(8 - 8)/2 = 0$ (zero) which implies that Ne_2 molecule is not able to exist.

Collective View of M.Os of Homonuclear Diatomic Molecules

For the purpose of comparison look at table (3.5) for M.O. pictures of Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2 and F_2 .

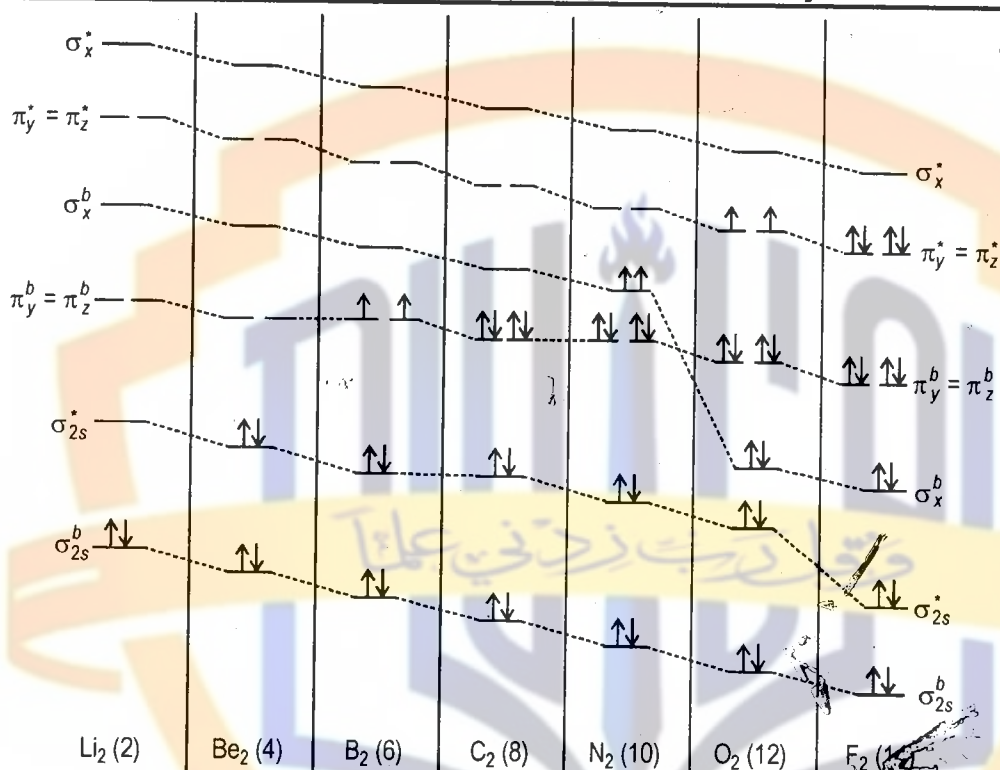


Fig. (3.41): Molecular orbital configuration of homonuclear diatomic molecules.

Formation of Molecular Orbitals in a Heteronuclear Diatomic Species of AB Type

Now, let us see how molecular orbitals are formed in a hetero-nuclear diatomic species of AB type. The examples are **CO**, **CN**, $\text{CN}^\ominus$, **NO**, $\text{NO}^\ominus$ etc. Here A and B atoms belong to 2nd period of the periodic table and electronegativity of A is less than that of B. Since both the atoms belong to 2nd period, each of them has 2s, 2p_x, 2p_y and 2p_z atomic orbitals (4 atomic orbitals) in its valence-shell. If atomic orbitals on atom A are represented (2s)_A, (2p_x)_A, (p_y)_A and (2p_z)_A and those on atom B are represented as (2s)_B, (2p_x)_B, (2p_y)_B and (2p_z)_B, the overlap of the four atomic orbitals on atom A with the corresponding four atomic orbitals of atom B gives eight molecular orbitals which are σ_{2s}^b , σ_{2s}^* , π_y^b , π_z^b , σ_x^b , π_y^* , π_z^* and σ_x^* . In a simplified way, the formation of these eight molecular orbitals can be shown as

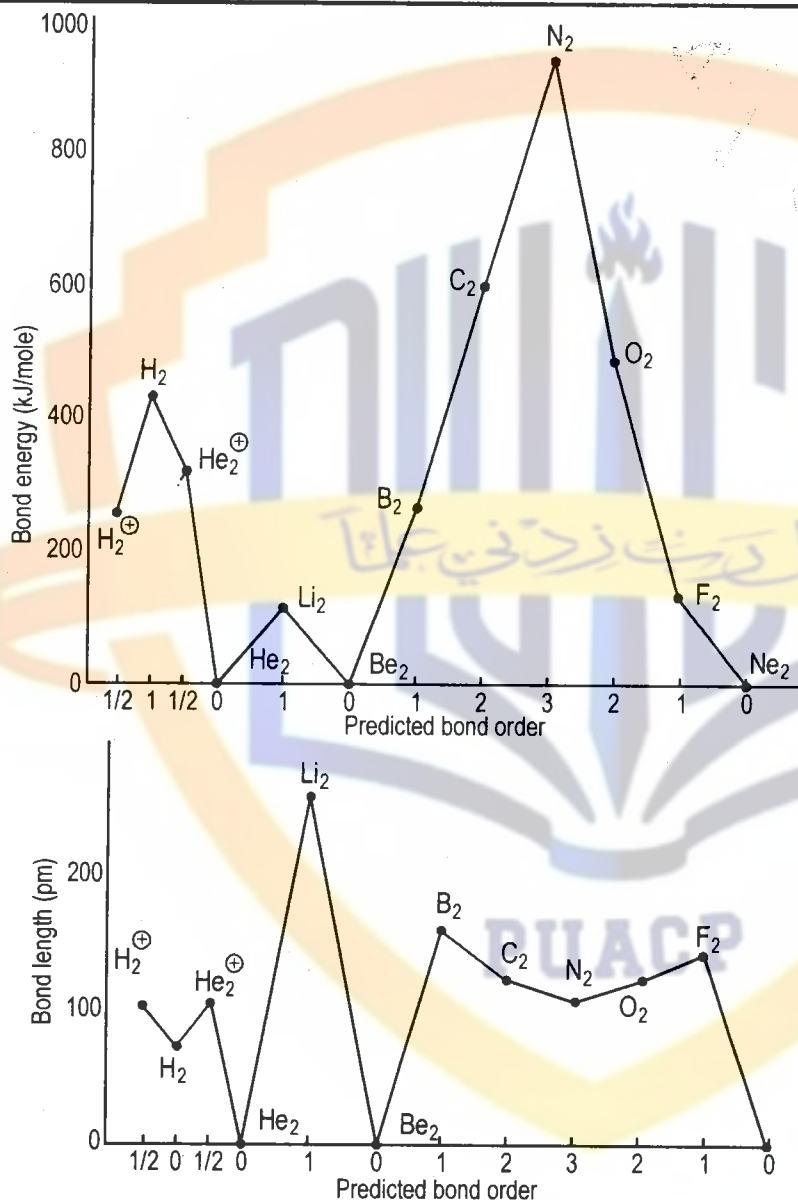


Fig. (3.42): Plot of bond energy and bond length against predicted bond order for first and second row homonuclear diatomic molecules.

theory provides us with a sound logical theoretical foundation to conceive polymeric molecular forms of elements.

Diatomic Molecules of the Periods Lying Lower to Second Period

There are several diatomic molecules and ions which belong to later periods. Experimental values for bond energies and bond lengths are known. Some values are given in Table (3.6) and graphical representations of some of them are shown in fig. (3.42)

Table (3.6): Bond properties of some homonuclear diatomics beyond period 2.

S. No.	Molecule	Bond Length (pm)	Bond Energy (kJ mol ⁻¹)
1	Na ₂	307.8	72.4
2	K ₂	392.3	49.4
3	Rb ₂	—	47.3
4	Si ₂	224.6	314.0
5	Ge ₂	—	272
6	Sn ₂	—	192.0
7	P ₂	189.3	477.0
8	As ₂	247.2	325.5
9	Sb ₂	221	298.3
10	S ₂	188.9	321.3
11	Se ₂	216.6	224
12	Te ₂	255.5	260
13	Cl ₂	198.8	239.2
14	Cl ₂ [⊖]	189.1	415
15	Br ₂	—	195
16	I ₂	266.7	148.7

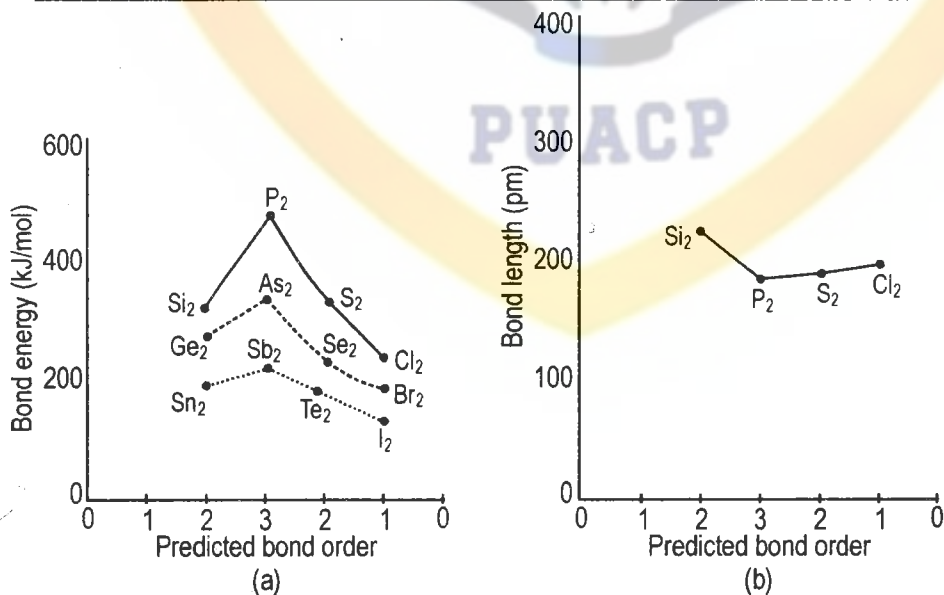


Fig. (3.43): Plot of bond energy and bond length against predicted bond order for elements beyond period 2.

3.6.10 Molecular Orbital Theory for Heteronuclear Diatomic Molecules

(1) MO Energy Level Diagram for Carbon Monoxide (CO) Molecule:

In CO molecule, the atomic orbitals of oxygen are lower in energy than the atomic orbitals of carbon. Due to this difference, the σ_{2s}^* orbital is higher in energy than the σ_{2px}^* orbital Fig. (3.44). Carbon monoxide possesses 10 valence electrons and, therefore, has a valence configuration as

$$(\sigma_{2s}^b)^2, (\sigma_{2px}^b)^2, (\pi_y^b = \pi_z^b)(\sigma_{2s}^*)^2$$

There are three net bonding pairs (one σ and two π) describing the $C \equiv O$ molecule.

We have placed two electrons of O-atom in $2p_z$, rather than $2p_x$. This is a matter of convenience for diagram. Actually $2p_x$, $2p_y$ and $2p_z$ are degenerate and are of equal status.

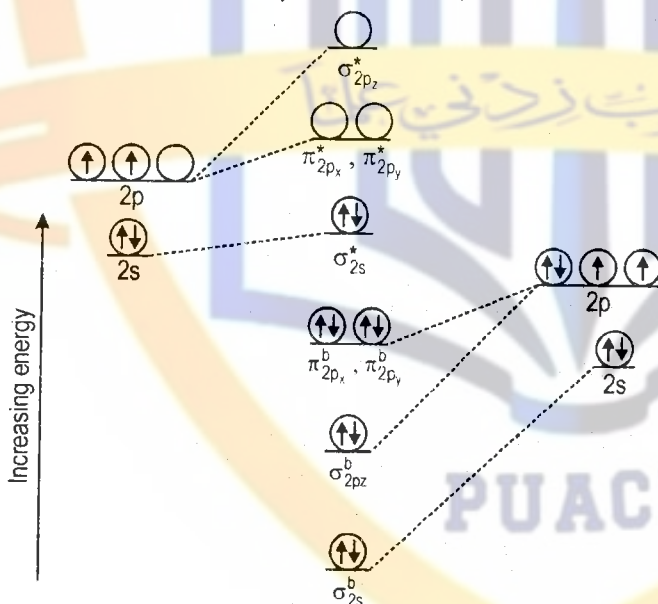


Fig. (3.44): Simplified molecular orbital diagram for the 2s and 2p atomic orbitals of the CO molecule.

(2) MO Energy Level Diagram for Hydrogen Fluoride (HF) Molecule:

The 2s atomic orbital of the fluorine atom does not interact with the 1s orbital of hydrogen atom because 2s is of low energy and is retained as atomic orbital. Only the $2p_x$ orbital of fluorine atom has proper symmetry and energy to interact with the 1s orbital of the hydrogen atom. In this way it gives one sigma bonding molecular orbital and one sigma anti-bonding molecular orbital Fig. (3.45). The $2p_x$ and $2p_y$ orbitals of fluorine yield non-bonding molecular orbitals at the same energies. Because of their symmetry they yield zero overlap with 1s orbital.

We are assuming that fluorine's other valence electrons remain localized on the fluorine. The 2p orbital of fluorine is shown at a lower energy than the 1s orbital of hydrogen on the diagram because fluorine binds its valence electrons more tightly. Thus the 2p electron on a free fluorine atom is at lower energy than the 1s electron on a free hydrogen atom. The diagram predicts that the HF molecule should be stable because both electrons are lowered in

energy relative to their energy in the free hydrogen and fluorine atoms, which is the driving force for bond formation.

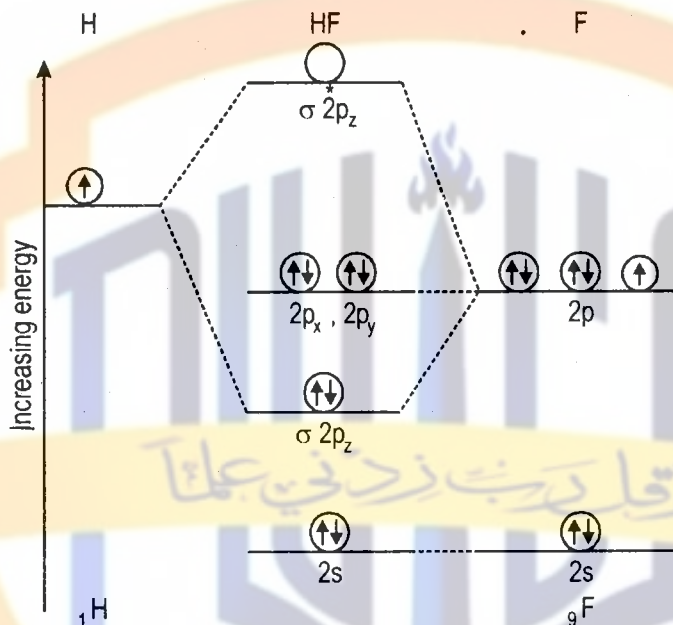


Fig. (3.45): The MO diagram for HF. HF is a polar molecule. In a polar covalent molecule, the MO diagram is asymmetric because the more electronegative atom has AOs of lower energy.

(3) MO Energy Level Diagram of HCl Molecule:

The energy of 3s and 3p atomic orbitals of the chlorine atom is lower than that of 1s atomic orbitals of the hydrogen atom. The 3s atomic orbital of chlorine atom does not interact with the 1s atomic orbital of H atom. The reason is that 3s has a very low energy and is, therefore, retained as atomic orbital. The 3p_z atomic orbital of chlorine atom has proper symmetry and energy to interact with the 1s atomic orbital of hydrogen atom. In this way it gives one sigma bonding molecular orbital and one sigma antibonding molecular orbital Fig. (3.46). The 3p_x and 3p_y orbitals of chlorine represent the two p orbitals on chlorine atom which lie perpendicular to the bond axis and hence do not overlap with 1s atomic orbital of the hydrogen atom.

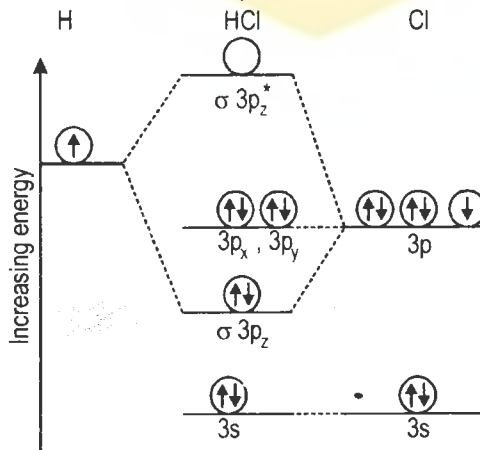


Fig (3.46): The MO diagram for HCl.

Molecular Orbitals of Boron Nitride (BN)

The molecule of BN is iso-electronic with the dinuclear C_2 molecule. Manage the energy level scheme for C_2 . It is expected that the molecular electronic configuration of BN will be as

$$KK (\sigma_s^b)^2 (\sigma_s^*)^2 (\pi_y)^2 (\pi_z)^1 (\sigma_x^b)^0$$

It seems that the bond energies of BN should be comparable to that of C_2 . Actually, the bond energy of BN is 385 kJ mol^{-1} while that of C_2 is 602 kJ mol^{-1} . There is a big difference. The reason is that π_y and π_z orbitals are very close to σ_x^b . The difference between this level and the π orbitals energy is comparable to the pairing energy of a pair of electrons in any of the π bonding orbitals. It is possible that one of the electrons from the π orbital is promoted to the σ_x^b orbital. The resulting molecular orbital configuration would change to

$$KK (\sigma_s^b)^2 (\sigma_s^*)^2 (\pi_y)^2 (\pi_z)^1 (\sigma_x^b)^1$$

Molecular Orbitals of BO, CN and $CO^\ominus$

The molecular species of BO, CN and $CO^\ominus$ are isoelectronic. They all possess nine valence electrons. The molecular electronic configuration can be represented as follows.

$$KK (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\pi_y)^2 (\pi_z)^2 (\sigma_x)^1$$

This energy level scheme shows that each of these species will have bond order of

$$\frac{7-2}{2} = \frac{5}{2} = 2.5.$$

Molecular Orbitals of CO , $CN^\ominus$ and $NO^\ominus$

These species are isoelectronic with molecular nitrogen (N_2). They all contain 10 electrons in their valence shell. Their energy levels are similar and the bond order for each species is 3.

$$KK (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\pi_{2p_y})^2 (\pi_{2p_z})^2 (\sigma_{2p_x})^2$$

The molecule NO possess 11 valence electrons. Its molecular electronic configuration is represented as follows.

$$NO = KK (\sigma_{2s}^b)^2 (\sigma_{2s}^*)^2 (\pi_y)^2 (\pi_z)^2 (\sigma_x)^2 (\pi_y^*)^1$$

The bond order is 2.5. The molecules have analogous structures as $O_2^\oplus$. Each have one unpaired electron, and are paramagnetic.

The bond length and bond energy data for these molecules are given in Table (3.7). This is for further critical assessment and configuration.

Table (3.7): Bond properties of some heteronuclear diatomic molecules.

S. No.	Molecule	Bond Energy (kJ mol^{-1})	Bond Length (pm)
1	C_2	602	124
2	BN	385	128

3	BO	545	120
4	CN	787	117
5	CO [⊖]	805	111
6	CO	1070	112
7	CN [⊖]	—	114
8	NO	678	115
9	NO [⊖]	—	106
10	N ₂	941	109
11	O	—	112

Comparison of VBT and MOT:

Similarities:

- Both theories
- are approximations.
 - predict the concentration of electron density between the nuclei.
 - interpret covalent bonds as orbitals embracing two atomic nuclei.
 - require that the combining atomic orbitals should be of nearly equal energy.
 - can account for directed valency.
 - predict the non-existence of helium molecule.

Dissimilarities:

Valence Bond Theory (VBT)	Molecular Orbital Theory (MOT)
(i) An electron moves under the influence of only one nucleus of an atom.	(i) An electron moves under the influence of two or more nuclei of a molecule.
(ii) VBT starts with individual atoms and considers the interaction between them.	(ii) MOT starts with nuclei of the constituent atoms.
(iii) In VBT, only half-filled orbitals of the valence shell take part in bonding and completely filled orbitals remain non-bonding orbitals.	(iii) In MOT, all the atomic orbitals, whether completely filled, partially-filled or vacant, overlap to form molecular orbitals, provided they have proper symmetry and comparable energy.
(iv) The atoms retain their individual identity.	(iv) The atoms lose their identity.
(v) VBT predicts the existence of anti-bonding orbitals in terms of repulsive forces.	(v) MOT predicts the formation of bonding as well as anti-bonding molecular orbitals.
(vi) VBT fails to explain the paramagnetic character of B ₂ and O ₂ molecules.	(vi) MOT can well explain the paramagnetic character of B ₂ and O ₂ molecules.
(vii) The concept of resonance plays an important role in VBT.	(vii) The concept of resonance does not play any role in MOT.

3.7 General Properties of Metals

We know that metals are good **conductors** of heat and electricity. They are **opaque** and have high **refracting power**. Metals have **high melting** and **boiling points**. They crystallise with high co-ordination numbers of 8 or 12. Such properties of metals cannot be explained on the basis of normal ionic or covalent bonding.

3.7.1 Nature of Bonding in Metals

The atoms of a **metal crystal** are identical. These cannot be bound together by ionic bonds. In ionic compounds cations and anions are bound together in an ionic crystal. Ionic bond is formed between atoms of **different electronegativity**. Van der Waals forces are too weak to account for high melting points of metals. Each atom in a metal crystal cannot be bonded to each of its **8 or 12 neighbours** by ordinary covalent bonds. Actually, in metals, the number of valence-electrons is **insufficient** for the formation of covalent bonds with all the close or closest neighbours.

It means that there is a new type of **cohesive force** in metals which is called **metallic bond**.

3.7.2 Theories of Metallic Bonds

How the metal atoms in a metal crystal are bonded together has been answered by the following theories:

(1) **Electron Gas Theory:**

This theory says that each atom in a metal crystal **loses all its valence-electrons**. The electrons thus obtained form an **electron pool or gas**. The positively charged metal ions are held together by this electron pool or gas. The positively charged metal ions **do not float randomly** in the sea of electrons. They have definite positions at measurable distances from each other in the crystal lattice. The valence-electrons belong to the crystal as a whole. They are free to move throughout the lattice from one part of the crystal to the other. These **electrons behave as gas molecules** which move freely throughout their container. The number of such free electrons is not certain.

Thus, according to this theory, metallic solids may be supposed as a collection of positive atomic cores immersed in a fluid of mobile electrons or sea of mobile electrons Fig (3.47). The force that binds a metal ion to the mobile electrons within its sphere of influence is known as metallic bond.

This theory was first proposed by **Drude in 1900** and was later on developed by **Lorentz (1916)**. This theory is also, therefore, called **Drude-Lorentz theory**.

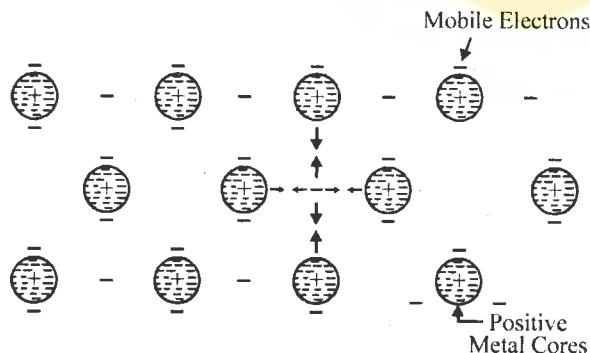


Fig. (3.47): Representation of sea of mobile electrons and positive metal cores.

Electron Gas Theory and Properties of Metals:

(i) Metallic Lustre:

When a beam of light falls on the surface of a metal, the **mobile electrons** move to higher energy levels. When **electrons come back** from the excited state to the ground state, in one or more jumps, the light of all wavelengths in the visible region is emitted. So when light falls on a metal surface, it appears as if light is being reflected. The surface, therefore, emits the typical metallic **lustre**.

(ii) Electrical Conductivity:

The electrical conductivity of metals is due to the presence of mobile valence-electrons. These electrons are free to move, and thus conduct electricity through the metal Fig. (3.48).

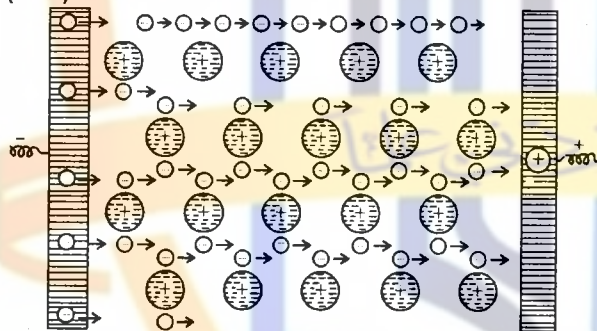


Fig. (3.48): Electron gas theory and explanation of conduction of electricity in metallic crystals on the basis of electron-pool theory.

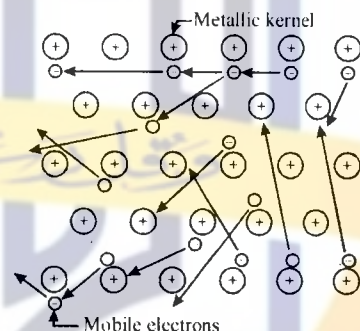


Fig. (3.49): Electron gas model and conduction of heat through metallic crystals.

(iii) Thermal Conductivity:

When one part of a metal is heated, the electrons in that part acquire a large amount of kinetic energy. **These electrons, being free, move rapidly through the crystal and conduct heat to the other parts of the metal.** This process is repeated and results in quicker heat transfer to the other end of the metal Fig. (3.49).

(iv) Malleability and Ductility:

Metals can be **beaten into their sheets** i.e., are malleable and can be **drawn into wires** i.e., are ductile, by very little expenditure of energy.

The force of attraction between $M^{\oplus}$ ions and the valence-electrons is uniform in all directions. So there are **no localised bonds**. It means that the bonds holding the crystal lattice in metals are **not rigid** as they are in covalent solids such as ice. This means that a layer of **kernel** can easily be shifted along a plane from one lattice site to the other one. This is done when a **shearing stress** is applied on a metallic crystal Fig. (3.50).

In the final state of the shifting process the relative positions of **kernels** remain the same as they were before shifting, since delocalised electrons are available everywhere. Thus we see that within the crystal each kernel has the same position with respect to its neighbours before and after shifting process and electron gas surrounds the kernel symmetrically in both the positions. **The ease with which new metal bonds may be formed by moving the metal ions from one lattice site to another is responsible for malleability and ductility.**

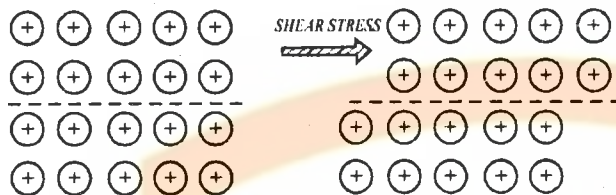


Fig. (3.50): A layer of metal ions slides past another layer when a shearing stress is applied on a metallic crystal.

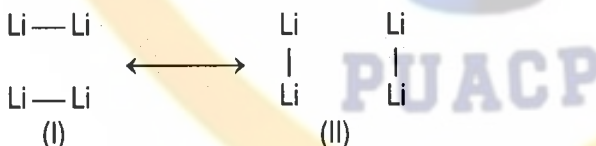
(v) **High Elasticity:**

As we have discussed that ions can change position without breaking the metallic crystal. It shows that under a stress, the structure is temporarily changed and attains the original position on removing the stress.

(2) **Valence Bond Theory:**

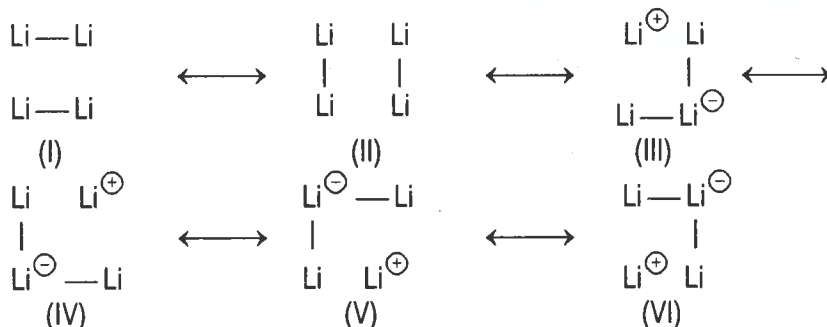
This theory of metallic bonding was introduced by **Pauling**. According to it the metallic bonding is **essentially covalent in origin**. The metallic structure involves **resonance of electron-pair bonds** between each atom and its nearest neighbours.

To make this theory understandable we take the example of **Li-metal**. In lithium metal each atom has **8 nearest neighbours** as shown by X-ray studies. The configuration of lithium atom is $1s^2, 2s^1$. It shows that lithium atom has only **one valence-electron**. This electron is insufficient to form electron-pair bonds between each atom of lithium and all its 8 neighbours. Consequently it is assumed that **resonance** takes place throughout the lithium solid. If each atom of lithium retains its single valence-electron, the resonance of pairs of bonds must be assumed to **synchronised throughout** the solid. Thus, in **two dimensions**, various resonance forms may be shown as follows:



Be careful in thinking that only four atoms have been shown in these forms but the actual bonding includes all of the atoms of the crystal and is three-dimensional.

To show the better resonance, we assume **unsynchronised resonance** between the arrangements involving ions. We get all the following **contributing structures**, which are relatively more stable:



These covalent bonds in III, IV, V, VI are resonating covalent bonds. The formation of these covalent bonds has been explained as follows.

In electronic configuration of lithium atom viz $1s^2, 2s^1 2p_x^0 2p_y^0 2p_z^0$ there are three vacant $2p$ -orbitals. The energy of these three- $2p$ -orbitals is not very different from that of the valence orbitals viz $2s$ -orbital. So an electron from $2s$ -orbital of one lithium atom can easily be transferred to one of three $2p$ vacant orbitals of the other lithium atom to convert it into $\text{Li}^\ominus$ ion with the configuration $1s^2, 2s^1 2p_x^1 2p_y^0 2p_z^0$. The lithium atom which is giving the electron from its $2s$ -orbital is converted into $\text{Li}^\oplus$ ion with the configuration $1s^2, 2s^0$.

The two vacant orbitals namely $2p_y^0$ and $2p_z^0$ can accept the conductivity electrons and thus contribute to metallic conduction. These two $2p$ vacant orbitals are termed as **metallic orbitals**.

Hence there should be orbitals, energetically not very different from the outermost occupied orbitals, in the configuration of the metallic atom for the formation of resonating covalent bonds.

(3) Molecular Orbital Theory–Band Theory:

This theory says that the electrons in a metal are considered to belong to the **crystal as a whole** and not to individual or any pairs of atoms. According to this theory metallic bonding results from the **delocalisation** of the free electron orbitals over all the atoms of a metal structure.

This is the **quantum mechanical treatment** of the metallic crystal and is similar to molecular orbital theory of covalent molecules.

When a diatomic molecule is formed, two atomic orbitals of equal energy combine together to give **two molecular orbitals**. One is **bonding molecular orbital** and one **anti-bonding molecular orbital**. Similarly when three atomic orbitals of a three-centric system combine, three molecular orbitals are obtained i.e., **bonding molecular orbital**, **anti-bonding molecular orbital** and **non-bonding molecular orbital**. If we have n -centric system where n is very large, we get n molecular orbitals.

Thus if a very large number of atoms of a metal, say 10^{23} , are brought together to form the metallic crystal, 10^{23} molecular orbitals are formed. The energies of these molecular orbitals are so closely spaced that they appear to be a **continuum**. This is which are called a **quasi-continuous energy band**. Hence the name **band theory**. These molecular orbitals are delocalised, i.e., these belong to the crystal as a whole.

Again consider molecular orbitals i.e., energy bands of Li which are formed in a lithium crystal. Electronic configuration of lithium atom is $1s^2, 2s^1, 2p^0$. If n atoms of lithium metal are allowed to combine to form lithium crystal, they will form the following three energy bands.

There are three types of orbitals as $1s$, $2s$ and $2p$. There will be three types of bands.

(i) **1s-band**. This band is formed by the combination of n $1s$ atomic orbitals of n lithium atoms and hence contains **n energy levels**. All of them are completely filled. Total number of electrons in this completely filled $1s$ -band is $2 \times n = 2n$. Since $1s$ orbital is the inner-orbital, it is screened from the influence of other atoms by valence electron. It does not appreciably interact and hence is narrow at equilibrium inter-atomic distance. $1s$ -band is the **non-conduction band**.

(ii) **2s-band**. $2s$ orbital are the valence-energy levels. These are very much influenced by the presence of other atoms and are quite spread out. This is called the valence-band.

Since each lithium atom contains only one 2s-orbital, n lithium atoms have n levels in 2s-band. These n levels of 2s-band have n electrons (each 2s-level has only one electron). As the maximum capacity of n levels of 2s-band is to contain $2 \times n = 2n$ electrons, 2s-band is half-filled. **The upper-half of this band is empty while the lower-half is completely filled.** The levels of 2s-band are so near in energy that the electrons from the lower-half can go to the upper-half by energy available even at ordinary temperature.

(iii) **2p-band.** p -orbital is **triply degenerate**. n atoms of Li solid will give $3n$ levels in 2p-band whose maximum capacity to contain the electrons is $2 \times 3n = 6n$ electrons. Since 2s and 2p levels are near in energy, 2s and 2p bands overlap and the electrons in 2s-band, therefore, have additional levels, differing very slightly in energy, to move about. The portion where 2s and 2p bands overlap is called **overlap zone**. This zone contains n electrons. **It is only 1/8th filled.** Actually the maximum capacity of quasi-continuous band formed from 2s and 2p bands to contain the electrons is of $2n + 6n = 8n$ electrons.

(4) Permitted and Forbidden Bands

Those electronic energy bands where the electrons can move are called the **permitted bands**. These are also called **Brillouin zones**. Two Brillouin zones are separated from each other by empty bands which are called **forbidden zones** or **forbidden energy gap**. For example, filled 1s-band which is a non-conducting band is separated from the half-filled 2s-band by an empty zone which is called forbidden zone or forbidden energy gap. The energy gap between 1s and 2s bands is quite high and hence the promotion of electrons from the lower non-conducting 1s-band to the higher conducting 2s-band is prevented. The width of a Brillouin zone depends on the overlapping of electron clouds. The level below which all energy levels are filled is termed **Fermi level** which may fall within a band or within the gap between the bands Fig. (3.51).

(5) Valence and Conduction Bands:

A solid crystal may be supposed to contain energy bands. Energy bands formed by the inner energy levels are very narrow while those obtained from the valence levels are quite wide and are known as valence bands. The valence bands which are completely filled are called **non-conduction bands** while those which are partially filled are called **conduction bands**. In conduction bands there is a large number of closely packed energy levels Fig. (3.44).

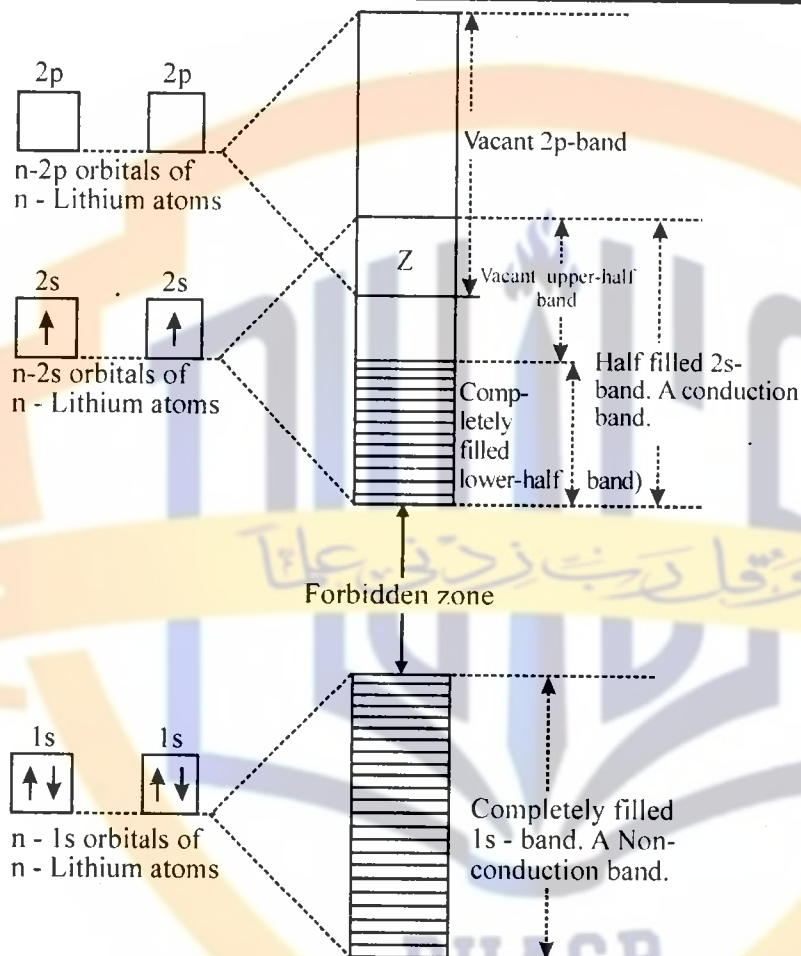


Fig. (3.51): Formation of 1s, 2s and 2p energy bands in lithium crystals. "Z" represents the overlap zone where 2s and 2p bands overlap.

(6) Physical Properties of Metals in the Light of Band Theory:

Now it is easy that we can explain why some solids are good conductors (metals) or electricity and others are non-conductors or **insulators** (non-metals) or **semi-conductors** (semi-metals).

(i) Conductors.

Those materials in which **plenty of electrons** are available for electric conduction are called conductors.

In terms of energy bands the conduction band of a conductor is partly filled and contains a very large number of electrons. These electrons can be promoted to the nearby vacant levels in the same band by electric or thermal energy. This explains the high electrical and thermal conductivity of the metals.

(ii) Insulators.

Those materials in which valence electrons are bound tightly to their parent atom and hence require very large electric current to remove them from the attraction of nuclei, are called insulators.

An insulator has **empty conduction band**. It means there are no electrons in this high energy band of the valence-shell. The energy gap between the valence-shell band which is completely-filled (**non-conduction band**) and the conduction band is very large Fig. (3.52). Thus energy required for shifting an electron from the completely-filled valence-shell band to the empty conduction band is very high and is, therefore, not normally available. This explains that **non-metals are insulators**.

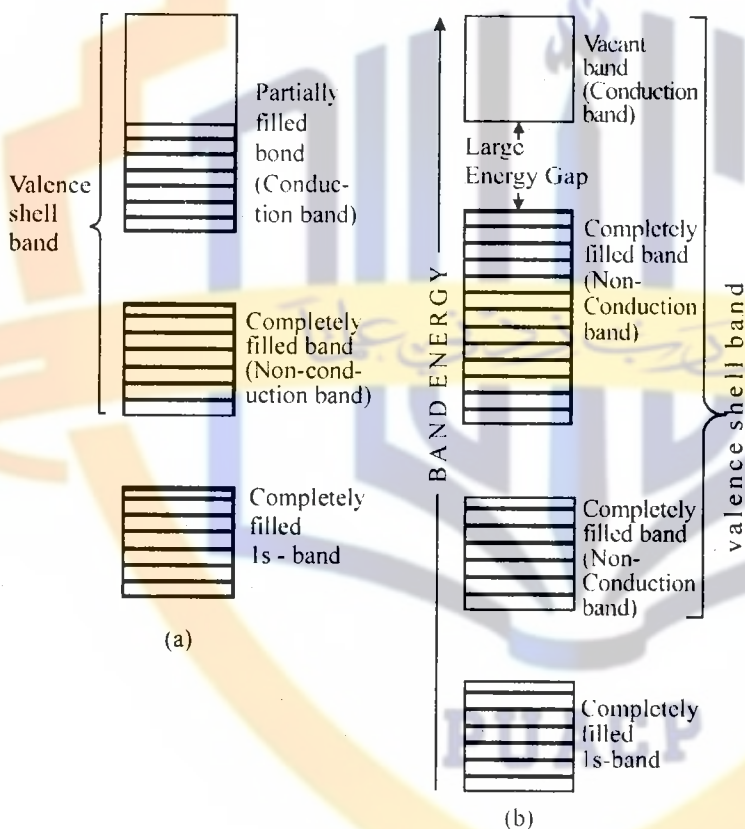


Fig. (3.52): Energy bands in a (a) conductor (metal) and (b) non-conductor or insulator (non-metal)

(iii) Semi-conductors. Those materials whose electrical properties lie between those of insulators and good conductors are called semiconductors. When these semi-conductors are in their extremely pure form, these are called **intrinsic semi-conductors**. **Germanium** and **silicon** are the common examples of intrinsic semi-conductors.

Semiconductors have filled valence band (non-conduction band) and empty conduction band. The forbidden energy gap between them is so small that even at room temperature many electrons from the filled valence-band jump to the vacant conduction band. **This shows low thermal conductivity.** As the temperature is increased, width of the forbidden energy gap is decreased and hence some more electrons jump to the conduction band. In other words the increase in temperature increases the conductivity.

Be careful!

In case of metallic conductors increase in temperature decreases their conductivity as the movement of electrons is interfered by greater vibrations of the kernels.

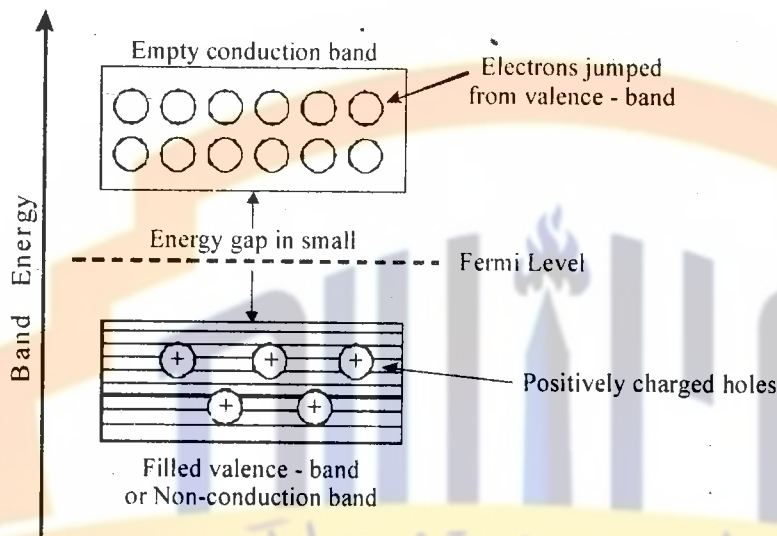


Fig. (3.53): Energy band diagram for an intrinsic semi-conductor at room temperature.

Now, when an electron is liberated into the conduction band, a **positively-charged hole** is created in the valence-band. Number of holes is equal to the number of electrons jumped to the conduction band. Energy band diagram for an intrinsic semi-conductor at room temperature is shown in Fig. (3.53). Here **Fermi-level** lies exactly in the middle of the energy gap. **Fermi level** is that level below which all energy states are filled.

EXERCISE

LONG QUESTIONS

VALENCE BOND THEORY

- The molecule of H_2 is formed from two H-atoms. Sketch the **diagram** for all the **attractive and repulsive forces** during bond formation of H_2 . Give the idea of **compro-mize distance** and **B.D.E.** with the help of a graph.
- (a) Give main points of **Valence Bond Theory (VBT)** of bonding. Describe the **theorems** on which VBT is based.
- (b) What do you mean by a **approximate wave function**?
- Discuss in detail the formation of H_2 molecule by **method of approximate wave functions**. How energy lowering takes place when two atoms come close to each other?

VALENCE SHELL ELECTRON PAIR REPULSION THEORY

- Write down the **main points of Valence Shell Electron Pair Repulsion (VSEPR)** concept. Discuss the **geometries of molecules** of following type:
 - AX_2E
 - AX_3E
 - AX_2E_2
- How **VSEPR concept** gives the geometry of following molecules?
 - $COCl_2$
 - PCl_5
 - SF_4
 - SF_6

6. Predict the **structures** of following molecules according to **VSEPR theory**.
 (a) BrF_3 (b) XeF_2 (c) TeF_5^- (d) XeOF_4

HYBRIDIZATION AND SHAPES OF MOLECULES

7. (a) Discuss the **geometries** of CCl_4 , SiCl_4 and SnCl_4 on the bases of **hybridization**. How these molecules differ from each other?
 (b) Compare the structures of NF_3 and PBr_3 keeping in view **sp^3 hybridization**?
8. (a) What is the **resemblance** of NH_3 and H_3O^+ from **hybridization** point of view for N and O-atoms? Sketch the diagrams.
 (b) Why all the **three bonds** of H_3O^+ are exactly similar although one of bonds is co-ordinate covalent?
9. (a) PCl_5 and SF_4 undergo **sp^3d hybridization**. Why these molecules do not have same bond angles? Discuss in detail.
 (b) Discuss the geometry of ICl_2^- ion by proposed **hybridization** of central atom.
10. The molecules as ClF_3 , XeF_2 and I_3^- have something common in them. Illustrate this by the concept of hybridization. In which respect do they differ from each other in geometry?
11. Discuss the **geometries** of following molecules by concept of hybridization.
 (a) SF_6 (b) IF_5 (c) XeF_4
 In what respects their geometries are different from each other?
12. (a) Give two examples of **sp^3d^3 hybridization** and explain their geometries.
 (b) Discuss the geometry of BF_4^- by proposing the type of hybridization of B atom.
13. Explain the structure of NO_3^- , SO_4^{2-} and ClO_4^- ions by **Valence Bond Theory (VBT)**.
14. Discuss the structure of CO_3^{2-} ion by **hybridization** of central carbon atom. How does it differ from NO_3^- keeping in view the basic principles of VBT?

MOLECULAR ORBITAL THEORY

15. (a) Give main points of **molecular orbital theory (MOT)**. What is the concept of linear combination of atomic orbitals (LCAO)?
 (b) Compare **bonding** and **antibonding molecular orbitals**.
16. Draw the pictures of σ , σ^* , π , π^* **molecular orbitals** for the bond formation in the following molecules.
 (a) H_2 (b) F_2 (c) O_2 (d) N_2
17. (a) Draw the **energy level diagram** of various molecular orbitals.
 (b) Give **rules** for filling the **molecular orbitals**.
 (c) What is **bond order**? How is it calculated?

18. Give the **molecular orbital picture** of H_2 , H_2^+ , He_2 and He_2^+ . Which of them is **non-existent**? Indicate the most stable and the least stable from them.
19. Sketch the diagram in which MO pictures of N_2 , O_2 and F_2 may be compared. Calculate their **bond orders** and **magnetic properties**.
20. (a) Compare the bond strengths of O_2 , O_2^+ and O_2^- . Which of these has **strongest bond**?
- (b) How **MOT** convinces us that F_2 is **diamagnetic** while F_2^+ and F_2^- are **paramagnetic**?
21. Sketch the **MO picture** of CO , NO_2 and HF molecules.

THEORIES OF METALLIC BONDING

22. Name **various theories of metallic bonding**. How electrons gas theory explain the
 - (i) **metallic lustre**,
 - (ii) **electrical conductivity**,
 - (iii) **thermal conductivity**,
 - (iv) **malleability and ductility**?
23. Explain the **VTB** for metals.
24. (a) What is **molecular orbital theory** or **band theory** for metals?
- (b) Give the details of **permitted and forbidden bands**.
25. Discuss some **physical properties of metals**.

SHORT QUESTIONS

VALENCE BOND THEORY

1. How **attractive and repulsive forces** between two bonded atoms decide the **compromise distance** or bond length in a molecule?
2. The **first trial wave function** for H_2 molecule is $\psi_1(a) \psi_2(b)$, but is **faulty**. Which is major fault in it?
3. The **second trial wave function** is based on the fact that electrons of two atoms are **indistinguishable**. Comment.
4. The possibility of **ionic character in covalent bond** of H_2 is given by which trial wave function?

VSEPR THEORY

5. CO_2 has four bonding electron pairs, while $BeCl_2$ has only two, but both are linear molecules. Comment.

6. The representation of structure of CO_3^{2-} is $\text{O}=\text{C}(\text{O}^-)_2$, but all the bond lengths are equal. Why?

7. All the bond angles of **formaldehyde** are **not 120°**, although carbon is sp^2 hybridized. Give reason.
8. The geometries of compounds of **Pb** and **Sn** in (**PbCl₂**, **SnCl₂**) are **V-shaped** when they show the valency of two with halogens but not linear like **BeCl₂**. Comment.
9. The bond angles of **NH₃**, **PH₃**, **AsH₃** decrease in the same order although hybridization of central atoms is same i.e., sp^3 . Give reasons.
10. The structure of **SF₄** is **distorted trigonal bipyramid**. Comment.
11. Justify the **T-shaped** structure of **BrF₃**.
12. Why $lp-lp$ repulsion $>$ $lp-bp$ repulsion $>$ $bp-bp$ repulsion?
13. Arrange the following molecules in order of increasing bond angle **AsH₃**, **NH₃** and **PH₃**. Give reasons.

HYBRIDIZATION AND SHAPES OF MOLECULES

14. **PCl₅** can be formed, but **NCl₅** cannot be. Give reasons.
15. How many lone pairs are there in the outermost orbital of Cl in **ClF₃**?
16. **IF₅** has only **one pyramide** not two. Justify it.
17. Indicate the species with **sp^3 hybridisation** of central atom.
 - (a) ClO_4^- (b) SO_4^{2-}
18. Pick up species with **sp^3d^2 hybridization**.
 - (a) XeOF_4 (b) IF_7 (c) BrF_5 (d) XeF_6
19. Give **shapes** and **hybridizations** of following molecules or ions:
 - (i) CS_2 (ii) BH_4^- (iii) Hg_2Cl_2 (iv) GaI_3
 - (v) Al_2Cl_5 (vi) ICl_2^+ (vii) PF_3Cl_2 (viii) NF_3
20. Give the **shapes acquired** by the following gaseous molecules.
 - (i) SO_3 (ii) XeF_2 (iii) SCl_2
21. Molecules as **CH₄**, **NH₃** and **H₂O** involve the same kind of **sp^3 hybridization**, yet they possess different geometries. Why?

MOLECULAR ORBITAL THEORY

22. With help of MOT indicate **paramagnetic nature** of **B₂**.
23. On the basis of MOT, explain that **N₂** molecule is **diamagnetic**, while **O₂** is **paramagnetic**.
24. How does the MOT account for bond order of **N₂⁺**, **O₂** and **O₂⁺**?
25. Write **MO configuration** for **O₂** and **O₂⁺** and which of them is **paramagnetic**?
26. On the basis of bond order, predict which of the following species is the most stable?

O₂⁺, **O₂** and **Ne₂⁺**
27. Draw MO diagram for **O₂⁺**, **O₂⁺** and **O₂²⁺** and give the value of bond order in each case.

28. Give molecular orbital configuration of O_2 , O_2^+ and O_2^- . Give order of stability with appropriate reasons.
29. Write the ground state electronic configuration of N_2^+ on the basis of MOT and calculate its bond order.
30. Explain why the ionisation energy of N_2 molecule is higher than that of NO molecule?
31. CO is diamagnetic. Justify it.
32. On the basis of MOT, explain the magnetic character of NO molecule. Also give its bond order.
33. The bond energy of NO is 623 kJ/mol, whereas that of NO^+ molecule ion is 1049 kJ/mol. Justify.
34. Explain why N_2 molecule is more stable than NO molecule?
35. With the help of molecular orbital model predict the bond order and magnetism of P_2 molecule.
36. The bond energy of NO is 623 kJ mol⁻¹ whereas that of NO^+ molecular 1049 kJ mol⁻¹. Justify.

GENERAL PROPERTIES OF METALS

37. Write down five general properties of metals.
38. Which theory of metals is also called Drude-Lorentz theory?
39. Thermal conductivity of metals is explained on the basis of electron gas theory. Justify it.
40. How metallic character of Li metal is justified on the basis of VBT?
41. What do you mean by permitted and forbidden energy gaps in metals?
42. How does a semiconductor changes its nature?
43. On which ground you suspect that covalent bond formation is unlikely in metals?
44. Which are those properties of metals which compel us not to propose ionic or covalent bonds in metals like other molecules?
45. The positively charged metal ions in a pure metal structure, do not float randomly. Comment upon this statement.
46. How do you justify the shining luster of metallic surfaces on the base of electron gas model of metals?
47. Metals are malleable and ductile. Comment upon this statement.
48. How valence bond theory of Pauling shows bonding in Li-metal?
49. What is concept of delocalization of electrons in molecular orbital theory of metals?
50. How many BMOs and ABMOs can be there and their energy gap one mole of pure Na metal as 23 g of it is there?
51. What do you mean by Brillouin zones?
52. Define conduction and non-conduction bands in metals.
53. How do you compare conductors and insulators in the light of band theory of metals?
54. How do you compare semi-conductors and intrinsic semi-conductors?